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Application of an improved GAN with a filter mechanism in small-sample bearing fault diagnosis☆

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ABSTRACT

In the field of bearing fault diagnosis, GAN are commonly used to address issues such as the scarcity of fault samples and data imbalance. In view of the instability in GAN training and its inability to achieve fault classification, this paper constructs a CSGAN-GP model by integrating the advantages of Conditional Generative Adversarial Network (CGAN), Semi-supervised Generative Adversarial Network (SGAN), and Wasserstein Generative Adversarial Network (WGAN-GP). This model is capable of directional generation and recognition classification of fault samples. Meanwhile, to further improve the quality of the generated samples, a filter mechanism is established in the discriminator (D) by setting a dynamic threshold associated with the loss function, and the model training strategy is dynamically adjusted based on the discrimination results, thereby enhancing the model's discriminative performance and improving fault classification accuracy. The model was trained and validated on the SDUST and XJTU-SY datasets to assess its sample-generation capability and fault-classification performance. Throughout training, it remained stable and controllable, while the filtering mechanism markedly improved generated-sample quality and diagnostic accuracy. The Euclidean distance between generated and real samples was 0.332, and the cosine distance was 0.00021, both the lowest among all compared models. Under scarce-sample conditions the model achieved fault-identification accuracy, recall and F1-score of 93.44 %, 93.49 % and 93.71 %, respectively, outperforming compared models. These results indicate that the model is well suited to bearing-fault diagnosis scenarios with sample scarcity and data imbalance.

1. Introduction

As a critical mechanical component, bearings are widely used in various types of equipment. Fault diagnosis, remaining useful life (RUL) prediction, and anomaly detection of bearings are essential for ensuring normal equipment operation and improving production efficiency. In recent years, with the rapid advancement of sensor technology and computational power, methodologies in related fields have shifted from traditional model-driven approaches to data-driven ones. In particular, deep learning models, by constructing multi-layer network architectures, are capable of automatically extracting features from raw data, thereby enhancing the efficiency and accuracy of methods in bearing-related research. Zhong [1] proposed a lightweight convolutional neural network (SLCNN) model based on a self-attention mechanism and transfer learning. By fine-tuning a pre-trained model, this approach addressed the issue of limited computational resources. Bastami [2] improved fault prediction accuracy by optimizing the mother wavelet, decomposition hierarchy, and node selection, and validated the superior predictive performance through accelerated life testing. Qiao [3] developed a generalized method for bearing fault diagnosis under variable working conditions using a multi-scale convolutional neural network with a two-stage attention mechanism (TSAMCNN), which enhanced diagnostic accuracy and noise robustness. Li [4] employed a semi-gradient temporal difference reinforcement learning algorithm to optimize convolutional kernels and biases, thereby achieving higher accuracy and computational efficiency in RUL prediction. Wen Zhao [5] proposed an anomaly detection method for main bearings in wind turbines based on a cascaded deep learning model (CNN-LSTM), which enabled earlier detection of abnormal states in the main bearing.

Although deep learning models have demonstrated outstanding

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performance in many applications, they primarily rely on large amounts of labeled data, and their effectiveness is highly dependent on data quality. However, in real world operational scenarios, particularly in the field of fault diagnosis, there is often a scarcity of fault data along with a problem of data imbalance [6], which significantly degrades the performance of model training. To address this issue, many scholars have employed Generative Adversarial Networks (GAN) to generate fault samples. For example, Zhang YH [7] adopted the CEAGAN method with an auxiliary classification module, which extracts time–frequency features via short-time Fourier transform to resolve the imbalance in rolling bearing fault data; whereas Ruan D [8] combined CNN and GAN, and introduced envelope spectrum error to improve traditional GAN, thereby generating bearing samples containing more fault features. Zhang YH [9] proposed a fault diagnosis method based on VAE-GAN sample augmentation and an FLCNN classification model to address the imbalance in rolling bearing fault data, effectively enhancing the model's generalization ability and diagnostic accuracy. Hao WS [10] improved GAN by integrating transfer learning to generate new samples, thus alleviating the scarcity of fault samples and significantly improving diagnostic accuracy. Li K [11] proposed an evolutionary algorithm-optimized conditional GAN (ECGAN) that utilizes adversarial learning to generate new samples similar to the real distribution, thereby improving diagnostic performance under sample imbalance. Moreover, Xu [12] proposed a GAN network that fuses an attention mechanism with global feature crossing by incorporating a spatial channel fusion attention module in the generator and adding global feature crossing and noise-resistant modules in the discriminator, effectively mitigating sample generation bias under data imbalance and enhancing the accuracy of bearing fault diagnosis. Although the aforementioned models have, to some extent, addressed the problem of sample scarcity, due to the inherent structural limitations of GAN, the model training process remains unstable, the training results uncontrollable, and fault classification unattainable; therefore, many scholars have made improvements based on GAN.

To address the issue of uncontrollable data generation, most research has focused on the CGAN model. Li [13] adopted a method combining CGAN with CNN, which improves the controllability of sample generation and fault diagnosis accuracy by generating a balanced dataset; Ahang [14] employed CGAN to generate fault data based on normal data, effectively compensating for the scarcity of bearing fault samples. Meanwhile, Huang [15] proposed an improved rAC-GAN that uses limited data to generate multi-type fault samples, thereby enhancing the generalization ability of wind turbine gearbox bearing fault diagnosis. In addition, Fan [16] combined resampling with conditional WGAN-GP and implemented rolling bearing data augmentation and intelligent diagnosis through a grayscale image CNN. Furthermore, Luo [17] adopted C-DCGAN to balance imbalanced data, significantly improving fault diagnosis accuracy and model generalization ability; and Irfan [18] utilized a CGAN with spectral normalization and adaptive adversarial noise injection along with an internal convolution feature extraction method, achieving high-quality sample generation and excellent noise robustness, thereby greatly improving bearing fault diagnosis performance.

To address the instability of training traditional GAN, some scholars have proposed the WGAN-GP method based on the Wasserstein distance and gradient penalty strategy to improve training stability. Zhou [19] proposed a method combining WGAN-GP with CNN-SVM, which significantly improved diagnostic accuracy and classification efficiency by augmenting the data and stabilizing the training using the gradient penalty strategy; Zhu [20] employed a WGAN-based fault diagnosis method, utilizing continuous wavelet transform to generate time–frequency images and adopting Bayesian optimized semi-supervised loss to enhance data generalization and feature extraction capability. Meanwhile, Zhang [21] proposed an enhanced WGAN method, which stabilized training by introducing spectral normalization and TTUR, and generated high-quality synthetic samples using gradient penalty, thereby achieving excellent performance on imbalanced fault diagnosis tasks; Tang [22] combined the strategies of grayscale image transformation, GAN, and CNN by first converting vibration signals into grayscale images and then generating additional training samples via GAN to resolve data imbalance issues; in addition, Fan [23] proposed a full attention WGAN (FAWGAN-GN) that, by incorporating gradient normalization and a full attention mechanism in the generator to focus on key features, effectively solved the sample imbalance problem.

To address the issue of the model's inability to perform classification, some scholars have adopted semi-supervised strategies that utilize unlabeled data for fault recognition and classification in situations with insufficient labeled data. Yang [24] proposed a semi-supervised SGAN method based on two-dimensional images and adaptive normalization, which solved the problem of insufficient labeled data by applying variational mode decomposition denoising and improved normalization operations; Li [25] adopted a semi-supervised GAN to convert time-domain signals into spectral signals, and combined spectral clustering withsoftmax to integrate limited labels, effectively improving fault recognition accuracy and clustering performance. Furthermore, Xing [26] proposed an improved semi-supervised generative adversarial network that enhanced the classifier's discrimination ability and convergence speed through adversarial learning between the generator and the classifier and an enhanced feature matching algorithm; and Ding [27] constructed a GAN-SSLadder classification model based on a semi-supervised ladder network and GAN, achieving a test set recognition accuracy of 95.5 % under conditions of data imbalance and label scarcity, thereby further improving the performance of rolling bearing fault diagnosis.

In summary, current models in this field typically address only individual aspects of GAN-related challenges, and few studies in bearing fault diagnosis simultaneously focus on both the quality of generated samples and classification performance. In light of the advantages of the aforementioned models, this paper proposes a semi-supervised conditional generative adversarial network with gradient penalty (CSGAN-GP) that integrates the strengths of CGAN and SGAN while incorporating a GP strategy. This model ensures stability in both the sample generation and fault classification processes. Moreover, by introducing a filter mechanism with a dynamic threshold associated with the loss function, the quality of generated samples is further enhanced, thereby improving fault classification accuracy. Finally, the SDUST and XJTU-SY datasets were used as validation sets to evaluate the sample generation capability and fault classification performance of the proposed model, CSGAN-GP-FP. The cosine distance between the samples generated by CSGAN-GP-FP and the real samples was only 0.000227, and the recognition rates on different training and validation datasets reached 93.21 %, 95.47 %, and 98.36 %, respectively. The results indicate that the model possesses good stability and controllability, is capable of generating high-quality fault samples, and demonstrates strong fault identification capability, making it suitable for bearing fault diagnosis scenarios characterized by data scarcity and imbalance.

2. Principles of the model

2.1. GAN

Generative Adversarial Networks (GAN) were first proposed by Goodfellow in 2014 [28]. They represent a class of deep learning models that generate new data through adversarial training between two neural networks. A GAN comprises two main components: a generator (G) and a discriminator (D). The training process is formulated as a game between these two networks, where the generator aims to produce samples that are as realistic as possible to "cheat" the discriminator, and the discriminator strives to accurately distinguish between real data and generated samples. Their mutual objective is to improve iteratively through continuous optimization.

As illustrated in Fig. 1, a random noise vector is input to the

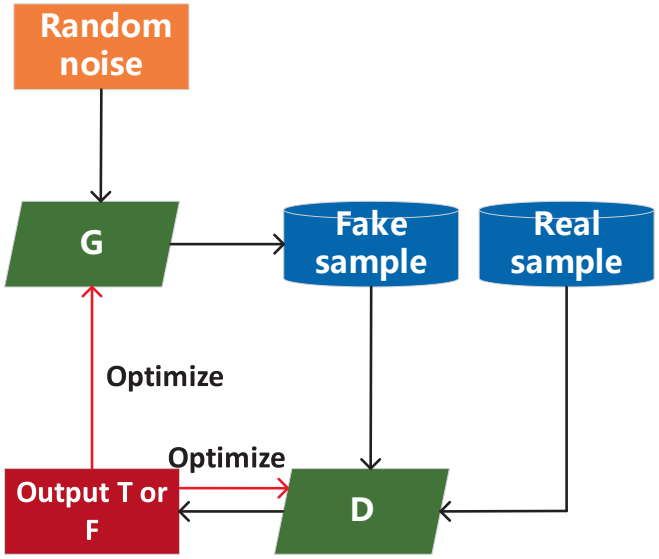


Fig. 1. GAN architecture.

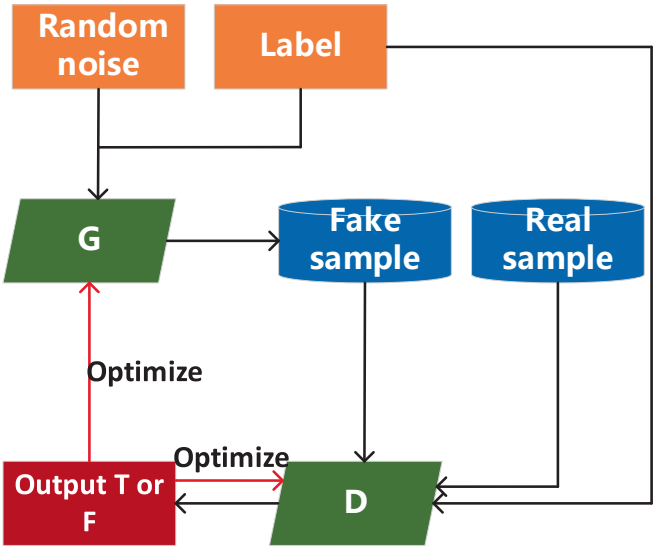


Fig. 2. CGAN architecture.

generator G to produce a generate samples. The generator's goal is to minimize the probability that the discriminator correctly identifies the generated sample as fake, thereby making the generated sample as similar as possible to real samples. The loss function for the generator is given by:

$$L_G = -\log D(G(z)) \tag{1}$$

where z represents the input noise vector, $G(z)$ denotes the generated sample, and $D(G(z))$ is the discriminator's estimate of the probability that the generated sample is real.

Subsequently, both real samples and generated samples are fed into the discriminator D, which outputs a probability value between 0 and 1. The discriminator's objective is to maximize the probability of correctly identifying real samples while minimizing the probability of misclassifying generated samples. Its loss function is defined as:

$$L_D = -[\log D(x) + \log(1 - D(G(z)))] \tag{2}$$

where x denotes real samples and represents the discriminator's estimation of the probability that x is real.

The optimization objective of GAN is to alternately optimize the loss functions of the generator and the discriminator, so that the samples produced by the generator become increasingly realistic while the discriminator becomes more accurate at distinguishing between real and generated samples, until the generator is capable of producing high-quality data that is nearly indistinguishable by the discriminator. The optimization problem of GAN can be described as:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z)))] \tag{3}$$

where p_{data} denotes the real data distribution and p_z denotes the noise distribution input to the generator. This adversarial process aims to achieve a Nash equilibrium between the generator and the discriminator, meaning that the samples produced by the generator become virtually indistinguishable from real samples.

2.2. CGAN

Conditional Generative Adversarial Networks (CGAN) are an extension of traditional GAN that introduce conditional variables to control the category or characteristics of the generated samples [29]. The structure is shown in Fig. 2 In contrast to standard GAN which rely solely on random noise, CGAN incorporates additional conditional information into both the generator and the discriminator, enabling the generator to produce samples with specific features, thereby further enhancing the controllability and diversity of data generation. By incorporating label information into both the generator G and the discriminator D, the loss function of the CGAN is given as:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}}[\log D(x|y)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z|y)))] \tag{4}$$

2.3. WGAN

Wasserstein Generative Adversarial Network (WGAN) is designed to address the common issues of gradient vanishing and mode collapse during GAN training [30]. By introducing the Wasserstein distance to measure the difference between the real sample distribution and the generated sample distribution, a smoother optimization process is achieved. Specifically, the Wasserstein distance is used in place of the traditional JS divergence as the metric to measure the distance between generated samples and real samples, and the loss function of WGAN is given as:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}}[D(x)] - \mathbb{E}_{z \sim p_z}[D(G(z))] \tag{5}$$

To ensure the effectiveness of the Wasserstein distance, the discriminator G needs to perform weight clipping to satisfy Lipschitz continuity. This process is typically enhanced by introducing a gradient penalty strategy (GP) to form WGAN-GP. By adding a gradient penalty term and generating interpolated samples between real and generated samples, the magnitude of the discriminator's gradient on these samples is ensured to be close to 1, thereby guaranteeing the network's representational capacity and stability. Its loss function is given as:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}}[D(x)] - \mathbb{E}_{z \sim p_z}[D(G(z))] + \lambda \mathbb{E}_{\hat{x} \sim p_{\hat{x}}}\left[\left(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1\right)^2\right] \tag{6}$$

where λ is the hyper-parameter for the gradient penalty, $\hat{x}$ is the sample generated by interpolating between real samples and generated samples, and $D(\hat{x})$ is the gradient of the discriminator with respect to the interpolated samples.

2.4. SGAN

Semi-supervised Generative Adversarial Networks (SGAN) combine the ideas of Generative Adversarial Networks and semi-supervised learning [31]. By utilizing unlabeled data for training, they are still

capable of generating high-quality samples and performing classification even when labeled data is insufficient. The structure is shown in Fig. 3, where category labels are introduced into the discriminator, enabling it not only to perform the real/fake discrimination task but also to undertake multi-class classification tasks. Therefore, the loss function of the discriminator comprises two parts: the loss for real/fake discrimination of generated samples and the loss for category prediction. The loss function of SGAN is given as:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{data}}[logD(x)] + \mathbb{E}_{z\sim p_z}[\log(1-D(G(z)))] - \mathbb{E}_{z\sim p_z}[\log P_y(y|G(z))] \quad (7)$$

where $P_y(y|G(z))$ represents the discriminator's category prediction for the generated samples.

CGAN, WGAN, and SGAN respectively make corresponding improvements to GAN for different issues. Some researchers have noted that although CGAN can generate conditional samples, its training process is prone to instability; by incorporating the GP strategy into CGAN [32], the stability of the targeted sample generation process is significantly improved. Other researchers, to address the problem that the model can only perform real/fake discrimination and cannot achieve fault classification, have combined the ideas of conditional generation and semi-supervised learning [33]. This model is able to more effectively achieve targeted sample generation and fault classification. In this paper, we combine the advantages of the two models, and, in order to further improve the quality of the generated samples, design a semi-supervised conditional generative adversarial network with gradient penalty (CSGAN-GP-FP) model based on a filter mechanism and the GP strategy.

3. Model design

The model structure is shown in Fig. 4, the operating principle of the model is as follows: the generator adopts the structural design of CGAN and WANG-GP, and by introducing conditional information and a gradient penalty strategy, it achieves controllability in sample generation as well as stability in the training process. In the discriminator component, the semi-supervised structure of SGAN and the design concept of CSGAN were adopted, replacing and extending the binary discrimination task into a multi-class classification task. The model directly uses the category prediction results as outputs for fault classification, thus eliminating the need to compute additional real/fake discrimination loss and reducing the training complexity. Especially under the condition of limited labeled data, the model can reduce the workload of data annotation while simultaneously performing sample generation and fault classification tasks. To further enhance the training effect and the quality of the generated samples, a filter mechanism is introduced, whereby sample filter is accomplished by comparing a dynamic threshold with the category prediction results, and the loss function is adjusted according to the filter results, thereby effectively filter out low-quality samples, enhancing the training effect, and improving fault classification accuracy.

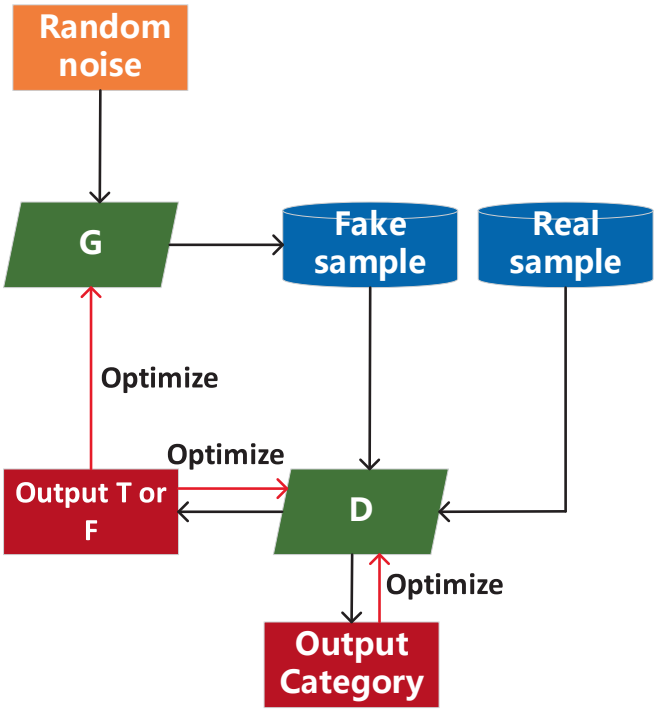


Fig. 3. SGAN architecture.

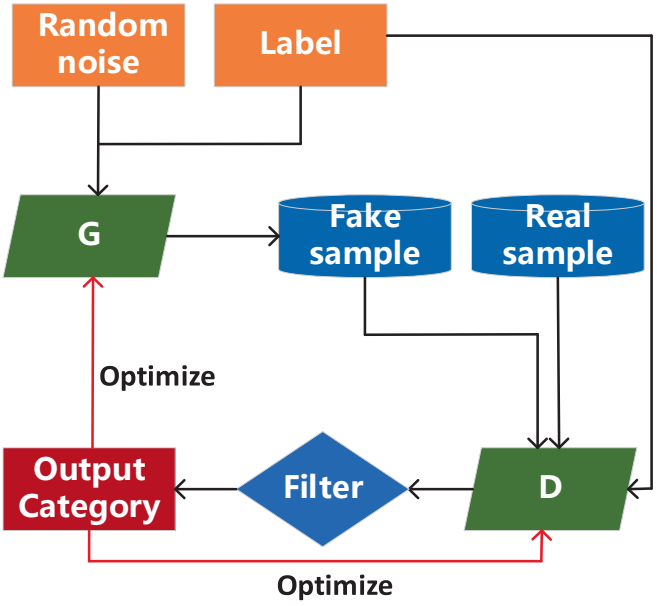


Fig. 4. CSGAN-GP-FP architecture.

3.1. Generator

3.1.1. Loss function

The generator part integrates the loss functions of CGAN and WGAN-GP, including the adversarial loss of the generator and the gradient penalty constraint loss on the discriminator, as shown in the following formula:

$$L(G) = -\mathbb{E}_{z\sim p_z}[\log D(G(z|y))] + \lambda \mathbb{E}_{\hat{x}\sim p_{\hat{x}}}\left[\left(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1\right)^2\right] \quad (8)$$

The first term ensures that the generated samples are as close as possible to real samples, increasing their "realism," while the introduction of the gradient penalty λ in the second term constrains the discriminator's gradient to be close to 1, thereby avoiding gradient vanishing or exploding during training and stabilizing the training process.

3.1.2. Model structure

The generator consists of an input layer, fully connected layers, a feature learning layer, and an output layer, as shown in Fig. 5.

First, a noise vector of length 100, randomly sampled, is concatenated with a conditional label of length N (encoded using one-hot encoding to represent fault categories), resulting in an input layer with 100 + N nodes. Then, the data is expanded to 256 dimensions through fully connected layer (FCL) and fed into the feature learning

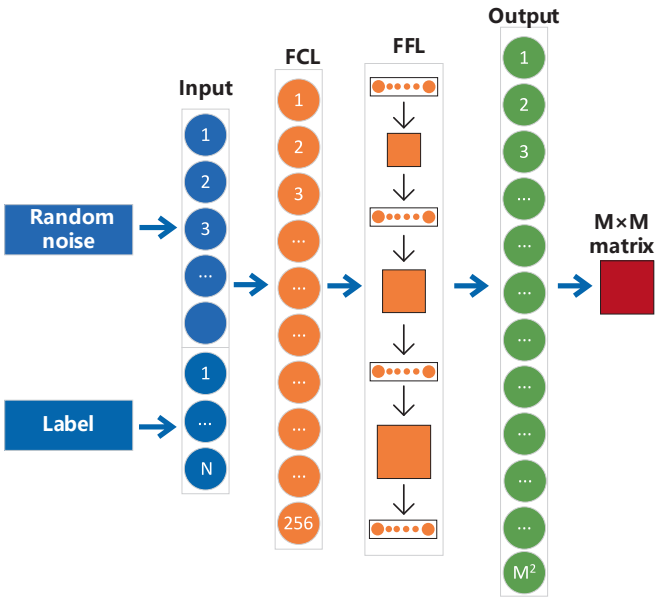


Fig. 5. Generator model architecture.

layer (FLL). After a series ofup sampling, convolutional layers, BN (Batch Normalization) layers, and LeakyReLU activation functions, the resolution of the generated samples is gradually increased while features are extracted, with batch normalization applied to the output. Finally, the output layer uses the Tanh activation function to reshape the data, yielding an output sample matrix of the specified size.

3.2. Discriminator

3.2.1. Construction of a dynamic threshold-based filter mechanism

Under conditions of data scarcity or data imbalance, the quality of generated samples is particularly important. If the generated samples fail to truly reflect the fault features, they may interfere with the training process and affect discrimination efficiency. Therefore, a filter mechanism is introduced to ensure that the generated samples meet certain quality standards, which can effectively improve classification accuracy and prevent the negative impact of low-quality generated samples on subsequent training.

Filter is accomplished by setting a dynamic threshold associated with the loss function. After the discriminator evaluates the generated samples, if the maximum probability value of a generated sample is lower than the dynamic threshold, that sample is excluded to avoid further processing of low-quality generated samples. The generated samples that meet the threshold condition will continue to participate in subsequent training, thereby improving the overall classification accuracy of the model. The dynamic threshold function is as follows:

$$\tau = \tau_{min} + (\tau_{max} - \tau_{min}) \times \frac{L_G}{L_G + L_D} \tag{9}$$

Here L_G and L_D are the loss functions of the generator and the discriminator, τ_{max} and τ_{min} represent the maximum and minimum values of the threshold, respectively, with the minimum value usually being the initial value. By adjusting the difference between the two, the training space can be modified. In the early stage of training, when the generator's capability is relatively weak and the quality of the generated samples is poor, setting a lower minimum threshold can ensure that the generator has sufficient room for learning, so that the generated samples are not screened out due to an overly strict threshold constraint. As the generator is optimized and training progresses, the quality of the generated samples gradually improves, the dynamic threshold gradually increases, and the filter mechanism's requirements for the generated samples also rise accordingly, ensuring that in the later stage of training only high-quality generated samples enter the fault classification task, thereby improving classification accuracy.

3.2.2. Loss function

The loss function for the discriminator part of CSGAN-GP comprises three components: the adversarial loss for real/fake discrimination, the type prediction loss, and the gradient penalty loss. After introducing the filter mechanism, the loss function becomes fourfold, with an additional filter penalty(FP) term γ added to the original components. Its computation formula is given as:

$$\gamma = \alpha \cdot \max(0, \tau - c) \tag{10}$$

where α is the hyper-parameter controlling the filter penalty term γ. If the discriminator's predicted probability for the generated sample's category is greater than the threshold ($\tau > c$), it indicates that the sample is not subjected to an additional penalty, its quality meets the standard, and its contribution to the discriminator loss remains normal, allowing it to participate in subsequent fault classification tasks; if the sample fails to pass the filter mechanism ($\tau < c$), the filter penalty term will reduce the contribution of that sample to the loss. The discriminator loss for CSGAN-GP with the filter mechanism is thus given as:

$$\begin{aligned} L_D = &-\mathbb{E}_{x\sim p_{data}}[logD(x|y)] - \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z|y)))] \\ &+ \lambda \mathbb{E}_{\hat{x}\sim p_{\hat{x}}}\left[\left(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1\right)^2\right] + \gamma \end{aligned} \tag{11}$$

3.2.3. Model structure

The discriminator mainly consists of a feature learning layer, a fully connected layer, and a filter layer, as shown in Fig. 6: Real samples and the samples generated by the generator G are taken as input and passed into the feature learning layer. They are processed through a series of convolutional layers, LeakyReLU activation functions, and pooling layers, where different sized convolution kernels and multi-channel operations are used for comprehensive, multi-level feature extraction. Then, the feature maps are fed into a fully connected layer for flattening, outputting an N-dimensional vector that represents the category prediction result. This output is passed through a Softmax activation function, and the maximum value of the result is then fed into the filter layer. Based on the result, it is determined whether the sample will participate in subsequent training, while simultaneously adjusting and providing feedback on the discriminator loss.

The structural parameters of generator and discriminator models are summarized in the Table 1.

3.3. Adaptive training strategy

In this paper, based on the filter mechanism, a training adjustment strategy is designed. Unlike traditional alternating training with fixed numbers of iterations, this strategy can adjust the alternating training between the generator and the discriminator based on the quality of the generated samples. Specifically, if the predicted category of a generated sample from the generator is greater than the preset threshold, it indicates that the generated sample has passed the filter, and the discriminator continues training; if it is lower than the preset threshold, it indicates that the quality of the generated sample has not met expectations, and the discriminator stops training, switching instead to generator training. This training strategy is based on the filtering results. Such a rationale-driven alternating training approach can reasonably allocate training resources and prevent overfitting, while effectively filtering out low-quality samples, stabilizing the training process, and significantly improving the accuracy of fault classification and the robustness of the model. The specific steps of the model training are shown in Fig. 7.

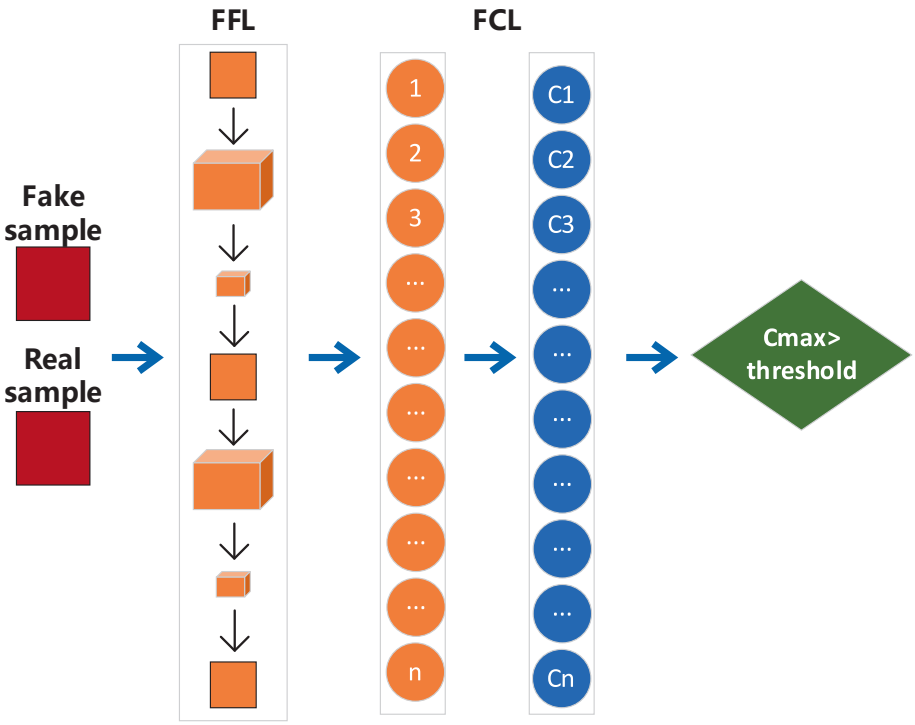


Fig. 6. Discriminator model architecture.

Table 1
Model architecture parameters.

Number of layers	Generator architecture	Discriminator architecture
1	FC Layer	Conv4_32
2	BN Layer	LeakyReLU Activation Layer
3	Up sample Layer	Pooling Layer
4	Conv3_128	Conv4_64
5	BN Layer	LeakyReLU Activation Layer
6	ReLU Activation Layer	Pooling Laye
7	Up sample Layer	Conv4_128
8	Conv3_64	LeakyReLU Activation Layer
9	BN Layer	Pooling Laye
10	ReLU Activation Layer	Conv4_256
11	Up sample Layer	LeakyReLU Activation Layer
12	Conv3_32	Pooling Laye
13	BN Layer	Conv4_1
14	ReLU Activation Layer	FC Layer
15	Up sample Layer	Filter Layer
16	Conv3_1	
17	Tanh Activation Layer	

(1) Initialize the parameters of the generator and the discriminator, and set the initial threshold τ_{min}, maximum threshold τ_{max}, gradient penalty term λ, and filter penalty term γ related parameters;
(2) Randomly sample a batch of noise data from the noise distribution and conditional labels (using one-hot encoding) as input to the generator;
(3) The generator G generates samples $G(z|y)$, which, together with real labeled samples, are used as input to the discriminator D to compute the loss;
(4) Fix the generator parameters and update the discriminator parameters accordingly;
(5) Let the discriminator train for five rounds to acquire better recognition capability;
(6) Input the maximum predicted category result c output by the discriminator into the filter layer and compare it with the threshold τ:
$c > \tau$, repeat steps (3), (4), and (5) until $c < \tau$;
$c < \tau$, input into D, and compute the loss of G;
(7) Fix the discriminator parameters and update the generator parameters accordingly;
(8) Repeat steps 1–7 until the iterations are finished.

4. Experiments and analysis

To comprehensively evaluate the model's performance, the experiments are divided into two stages. The first stage verifies the model's sample generation capability, while the second stage assesses its fault identification ability. Both stages are validated using the SDUST and XJTU-SY datasets, respectively. To ensure the model's applicability in real-world operational scenarios, noise is added to both datasets to test the model's robustness and generalization ability. The approach involves adding Gaussian noise to the signals and performing normalization processing to preserve the overall data trends without distorting the original fault features.

Gaussian noise is one of the most commonly used types of noise, and its mathematical expression is:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{12}$$

where $\mu = 0$ and σ denotes the standard deviation, representing the intensity of the noise. The following equation represents the addition of noise $n(t)$, generated from a Gaussian distribution, to the original signal $s(t)$ at time t, thereby forming the one-dimensional time series data $x(t)$ used for subsequent input and transformation.

$$x(t) = s(t) + n(t) \tag{13}$$

4.1. Validation on the SDUST dataset

The SDUST dataset originates from the fault implantation test bench for rotating machinery transmission components at Shandong University of Science and Technology. It is a standard dataset widely used in

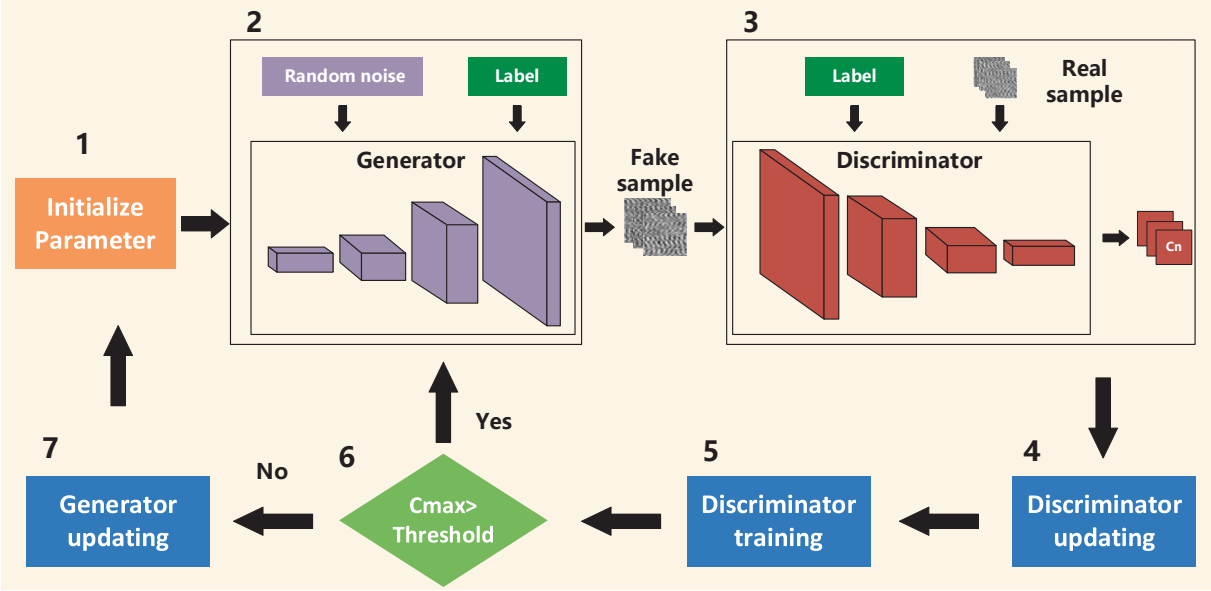


Fig. 7. Model training steps.

the field of fault diagnosis [34]. As shown in Fig. 8, the test bench mainly consists of an AC motor, a motor speed controller, test bearings, a gearbox, as well as a handle and a magnetic powder brake for load control. It enables fault diagnosis experiments for various rolling bearings and gears under different operating conditions.

This study utilizes a 12-class dataset of bearings collected at a sampling frequency of 25.6 kHz. The dataset includes the normal operating condition of the bearing as well as fault data of the rolling element, inner race, and outer race under various fault sizes, rotational speeds, and loads, which comprehensively reflect the fault characteristics of bearings under different working conditions. The specific fault types and corresponding labels are shown in Table 2.

To fully leverage the advantages of GAN-based image generation, this study converts one-dimensional fault signals into two-dimensional grayscale images. The specific steps are as follows:

(1) Continuously collect 1024 data points as one sample, and expand the sample set and improve image resolution through linear interpolation and sliding window processing;
(2) Reshape the processed signal sequence into a 64 × 64 two-dimensional matrix in a row-major order to facilitate subsequent mapping conversion;
(3) Referencing the Wen L's method [35], normalize each element in the two-dimensional matrix to the range [0,255] to convert it into an 8-bit grayscale image, with the formula as follows:

Table 2
Parameters of the SDUST 6502 bearing dataset.

Location	Label	Fault size (mm)	Rotational speed (rpm)	Load (N)
Normal	NO-1000-20	–	1000	20
	NO-1500-40	–	1500	40
	NO-2000-60	–	2000	60
Roll	RF-0.2-1000-20	0.2	1000	20
	RF-0.4-1500-40	0.4	1500	40
	RF-0.6-2000-60	0.6	2000	60
Inner	IF-0.2-1000-20	0.2	1000	20
	IF-0.4-1500-40	0.4	1500	40
	IF-0.6-1000-60	0.6	2000	60
Outer	OF-0.2-1000-20	0.2	1000	20
	OF-0.4-1500-40	0.4	1500	40
	OF-0.6-2000-60	0.6	2000	60

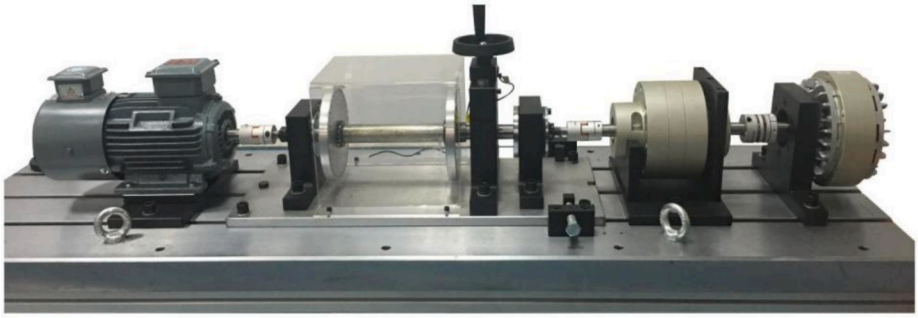


Fig. 8. Bearing fault test bench.

$$Q(i,j) = \text{round}\left[\frac{X(i-1)\times N + j - min(X)}{\max(X) - \min(X)} \times 255\right] \quad (14)$$

where $Q(i,j)$ is the value in the ith row and jth column of the two-dimensional matrix,N is the length and width of the grayscale image;X is the sampled value after processing the one-dimensional time series signal, the number of which is N^2. Compared with one-dimensional signals, two-dimensional grayscale images are advantageous for convolutional layers to fully learn the spatiotemporal correlation features of the signal and improve the quality of generated samples. Fig. 9 shows the time-domain waveforms corresponding to the 12 types of signal data, where it is clearly observed that different types of data exhibit distinct fluctuation patterns. Fig. 10 presents both the time-domain waveforms and their corresponding grayscale images, with the grayscale images used as the input for model training.

The model training environment was Python 3.9. The training input consisted of a noise vector with a dimension of 100 and 12 indicator labels. The transformed two-dimensional grayscale images were used as real samples. The batch size was set to 64, the learning rate (lr) was 0.01, and the number of training epochs was 2000. The Adam optimization algorithm was employed.

To evaluate the model's sample generation capability, this study conducted comparative experiments involving GAN and its improved variants: AC-GAN, DC-GAN, CGAN-GP, and CSGAN-GP-FP. GAN and CGAN-GP served as baselines to assess the effects of gradient penalty (GP) and filtering mechanism (FP) on sample generation performance. AC-GAN [36] introduces an auxiliary classifier based on GAN, while DC-GAN [37] applies convolutional network structures in both the discriminator and generator, and both are commonly used improved variants of GAN, serving as comparison models to validate the effectiveness of the proposed model.

Fig. 11 presents the comparison of generation loss curves among the five models. As shown in the figure, GAN, DC-GAN, and AC-GAN exhibit relatively high initial generation loss, while CGAN-GP and CSGAN-GP-FP start from zero. This is because GAN, DC-GAN, and AC-GAN are not initially constrained by gradient penalty, and their generators rely entirely on random noise to deceive the discriminators. As training progresses, the generator and discriminator continue to compete, leading to a gradual decrease in generation loss. Compared with GAN, DC-GAN benefits from its convolutional structure, resulting in a faster decline in generation loss and a relatively lower loss level. However, its overall generation loss remains higher than that of AC-GAN. AC-GAN introduces additional category information through an auxiliary classifier, which enhances the upper bound of loss reduction and enables the generator to maintain a lower generation loss during the middle and later training stages. Nevertheless, its final loss is still higher than that of the two models incorporating "penalty" mechanisms.

In contrast, CGAN-GP and CSGAN-GP-FP, benefiting from the penalty strategy, exhibit lower generation losses than the first three models. During the initial training phase, both models show significant fluctuations and increases in their loss curves. This is because, under the constraint of the penalty, the generator has limited knowledge of the data distribution while attempting to "cheat" the discriminator. As training progresses, the generator gradually learns the distribution characteristics of real samples, and the loss curve exhibits a downward trend and gradually stabilizes, showing slight fluctuations. The two models have nearly identical structures, with the key difference being that CSGAN-GP-FP additionally incorporates a semi-supervised structure (for subsequent fault classification) and a filtering mechanism. Ultimately, the generation loss of CSGAN-GP-FP decreases and stabilizes around 7, while that of CGAN-GP remains around 11. This indicates that the introduction of the filtering mechanism can effectively reduce generation loss, thereby enhancing the model's ability to capture the features of real samples.

To quantify the similarity between the generated samples and the real samples, this study adopts Euclidean distance and cosine distance as

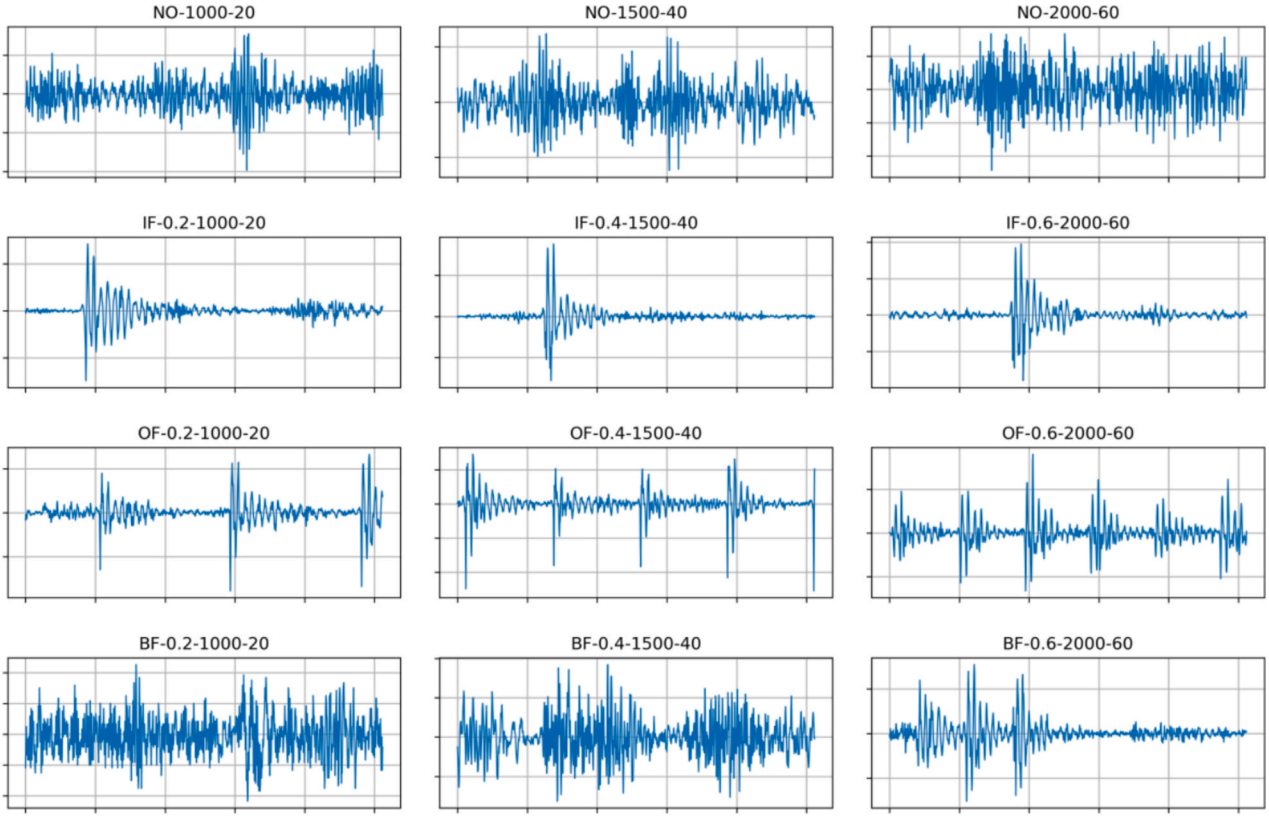


Fig. 9. Time-domain waveforms of signals.

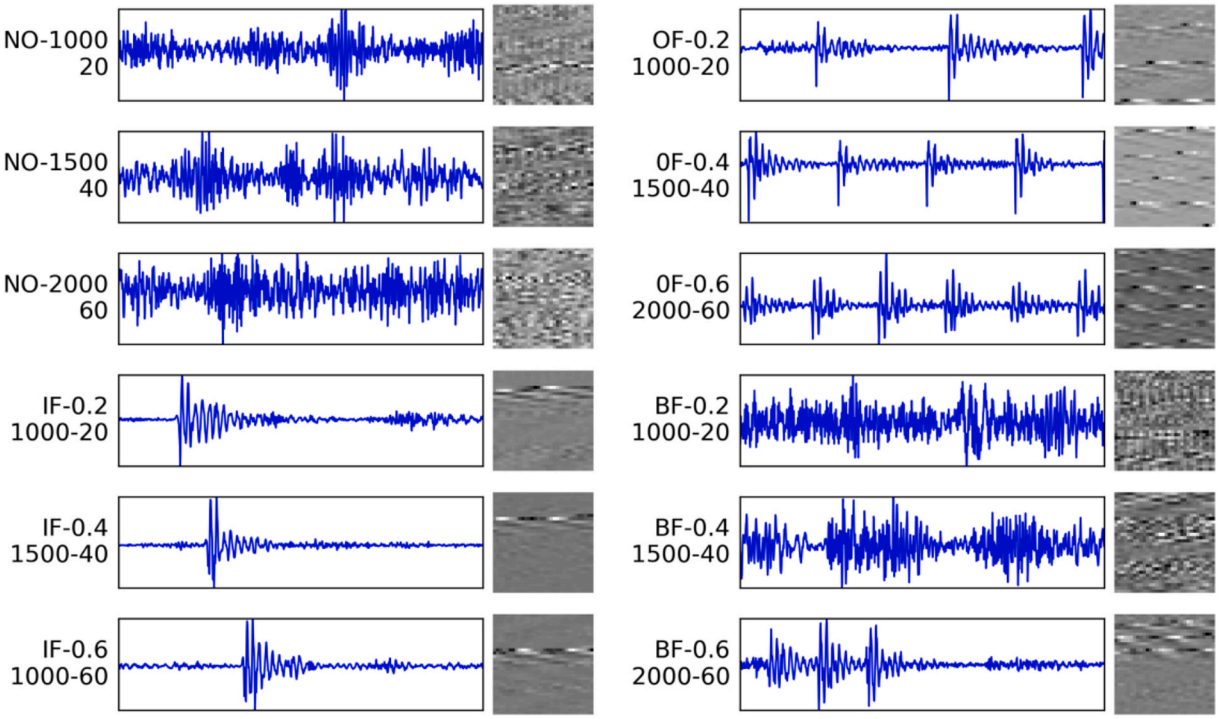


Fig. 10. 2D grayscale image conversion.

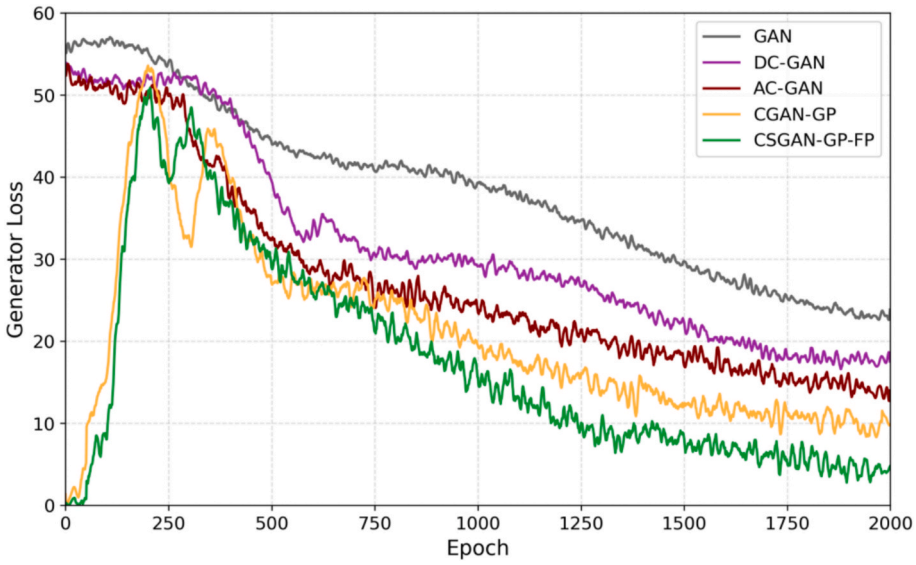


Fig. 11. Generation loss variation curve.

evaluation metrics. Euclidean distance measures the straight-line distance between two samples in the feature space, with smaller values indicating higher similarity between the generated and real samples in the feature space. Cosine distance, on the other hand, reflects the structural similarity between the compared objects, where smaller values represent greater structural similarity. Table 3 presents the calculation results for each model. It can be observed that, among the ten types of samples, the Euclidean distance and cosine distance corresponding to the proposed CSGAN-GP-FP model are 0.322 and 0.00021, respectively—the smallest values among all compared models. This indicates that the samples generated by CSGAN-GP-FP exhibit the highest structural similarity to real samples in the feature space, thereby demonstrating the superiority of the proposed model in terms of sample generation quality.

Fig. 12 presents the marginal spectrum plots of the real samples and the samples generated by CSGAN-GP-FP, with four types of samples selected for illustration: NO-1000-20, IF-0.2-1000-20, OF-0.2-1000-20, and BF-0.2-1000-20. As shown in the figure, the peak amplitudes in the marginal spectra of the real samples can be correspondingly observed at the same frequencies in the generated samples, and the overall trends of the two are generally consistent. Specifically, four frequency points are randomly selected, and the amplitude values at corresponding frequencies for real and generated samples are annotated separately. The amplitude values for the real samples are 0.005, 0.245, 0.006, and 0.348, while those for the generated samples are 0.025, 0.273, 0.017, and 0.375. At the same frequency points, the amplitude

Table 3
Comparison of euclidean distance and cosine distance calculation results.

Label	CSGAN-GP-FP		CGAN-GP		AC-GAN		DC-GAN		GAN	
	ED	CD	ED	CD	ED	CD	ED	CD	ED	CD
NO-1000-20	0.332	0.00022	0.371	0.00027	0.402	0.00030	0.438	0.00038	0.546	0.00055
NO-1500-40	0.318	0.00021	0.356	0.00025	0.387	0.00029	0.417	0.00034	0.579	0.00057
NO-2000-60	0.306	0.00020	0.348	0.00024	0.390	0.00032	0.458	0.00039	0.602	0.00074
RF-0.2-1000-20	0.361	0.00028	0.358	0.00029	0.402	0.00034	0.443	0.00041	0.509	0.00053
RF-0.4-1500-40	0.335	0.00023	0.372	0.00028	0.418	0.00033	0.462	0.00042	0.498	0.00056
RF-0.6-2000-60	0.304	0.00019	0.394	0.00026	0.384	0.00030	0.441	0.00037	0.631	0.00076
IF-0.2-1000-20	0.291	0.00018	0.331	0.00024	0.372	0.00029	0.425	0.00036	0.517	0.00054
IF-0.4-1500-40	0.370	0.00027	0.392	0.00030	0.428	0.00035	0.413	0.00043	0.488	0.00046
IF-0.6-1000-60	0.285	0.00017	0.343	0.00025	0.355	0.00027	0.438	0.00038	0.548	0.00054
OF-0.2-1000-20	0.297	0.00018	0.341	0.00023	0.368	0.00028	0.415	0.00035	0.506	0.00052
OF-0.4-1500-40	0.337	0.00022	0.352	0.00026	0.391	0.00031	0.447	0.00039	0.541	0.00055
OF-0.6-2000-60	0.326	0.00020	0.345	0.00025	0.377	0.00030	0.432	0.00037	0.526	0.00050
Age	0.322	0.00021	0.356	0.00026	0.380	0.00031	0.436	0.00038	0.541	0.00057

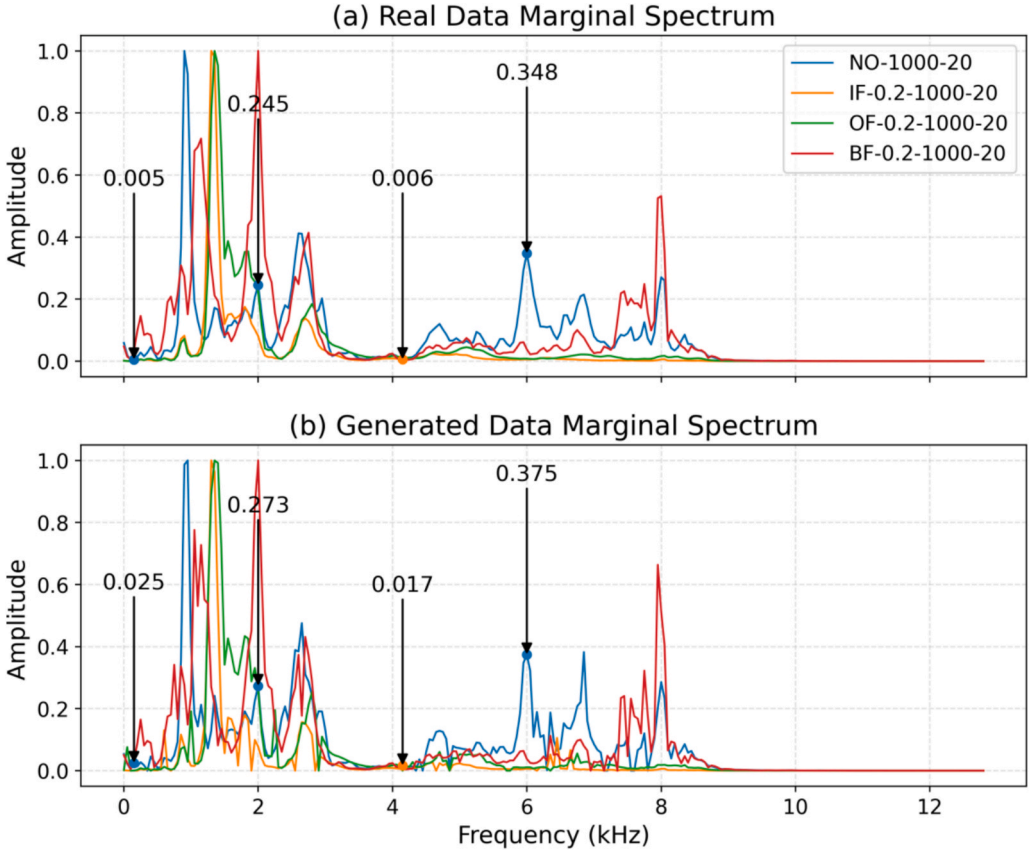


Fig. 12. True sample and generated sample marginal spectrum.

differences between the two are 0.02, 0.028, 0.009, and 0.027, respectively, all of which are less than 0.05. This indicates that CSGAN-GP-FP can effectively capture the characteristics of real samples and generate approximate samples, thereby producing high-quality approximate samples.

To further validate the impact of the filtering mechanism on the model's generation capability, a comparison was conducted between CGAN-GP and CSGAN-GP-FP. Samples generated by both models at training epochs 500, 1000, 1500, and 2000 were selected, and t-SNE feature analysis plots were drawn to visualize the distribution of generated and real samples.

As shown in Figs. 13 and 14, the distributional differences between the generated samples and the real samples gradually diminish with increasing training iterations, indicating that the features of the generated samples become increasingly similar to those of the real samples as training progresses. Further comparison reveals that at 1500 iterations, the feature distribution regions of the generated and real samples by CSGAN-GP-FP nearly overlap entirely, whereas CGAN-GP shows substantial but not complete overlap at the same stage. It is only at 2000 iterations that CGAN-GP achieves near-complete overlap, while CSGAN-GP-FP has already reached full overlap and even exhibits specific sample point fitting. This demonstrates that CSGAN-GP-FP can generate samples with higher similarity to real data in a shorter training period. It further indicates that the introduction of the filtering mechanism effectively enhances the model's learning efficiency, enables thorough learning of sample features, and ensures the stable output of high-quality samples.

In summary, CSGAN-GP-FP demonstrated good stability and

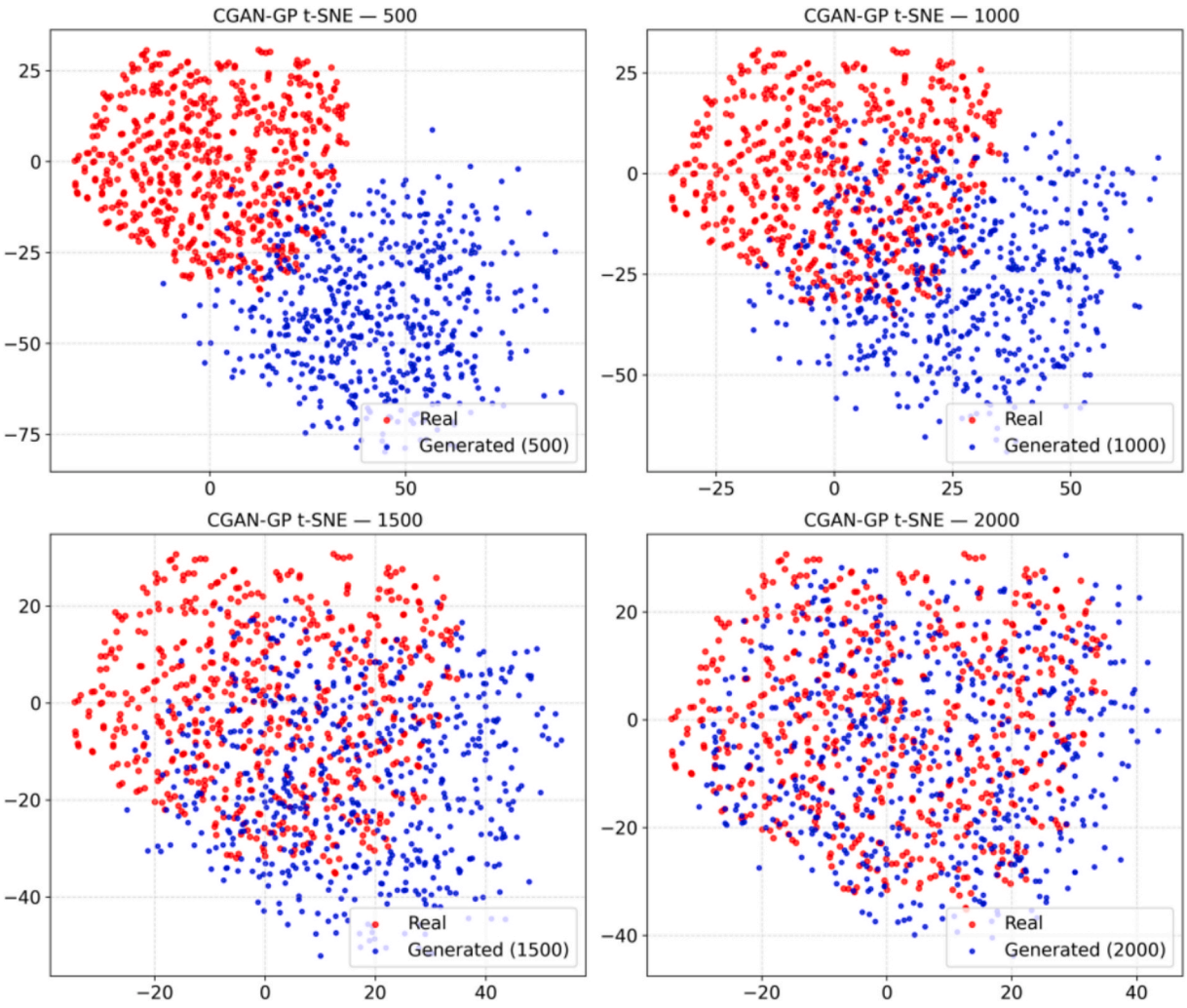


Fig. 13. T-SNE feature map of CSGAN-GP samples.

controllability during the training process, which improved the quality of the generated samples and provided a solid foundation for subsequent fault diagnosis.

4.2. Validation on the XJTU-SY dataset

Next, the fault identification capability of the model is validated using data from the Bearing Accelerated Life Test Rig at Xi'an Jiaotong University [38]. As shown in Fig. 15, the test rig is composed of an AC motor, a motor speed controller, a shaft, supporting bearings, a hydraulic loading system, and the test bearings, and it contains full life-cycle vibration signals of 15 rolling bearings under 3 different operating conditions.

In this study, 5 types of fault data from the XJTU-SY dataset are used as the validation objects. The data were collected via a horizontal accelerometer mounted on the shaft test rig, with a bearing rotational speed of 2100 r/min and a radial force of 12 kN. The fault data and corresponding labels are shown in the Table 4.

The 5-class fault dataset is divided into training and testing sets according to the ratios 2:8, 3:7, and 4:6, with 2000 iterations. The Fig. 16 shows the recognition rate curves as a function of the number of iterations.

First, considering the overall trend of the three curves, the recognition rates of the three curves are all around 20 % in the initial stage, which is due to the model being in the initial stage of training with random initialization; the generator produces low-quality samples and the discriminator fails to capture effective fault features, so the recognition results are equivalent to random guessing. As the number of iterations increases to between 150 and 400, the parameters of the generator and discriminator are updated through feedback, enhancing the model's feature learning capability. During this stage, the quality of the generated samples improves significantly, and the classification ability of the discriminator is strengthened, leading to a sharp increase in the recognition rates of the model under all three data ratios. Subsequently, after sufficient parameter updates, the model enters a stable convergence phase and the recognition rate gradually stabilizes; at this point, the model has a good grasp of the fault features, and as the number of iterations increases, the recognition rate shows a slight, steady upward trend.

At the same time, the recognition rate exhibits some differences due to the varying training set ratios. Among them, the recognition rate curve for the 4:6 ratio rises rapidly to a high level within the first few hundred iterations, whereas for the 3:7 and 2:8 ratios the rise in recognition rate occurs later, and the final recognition rates for the 3:7 and 2:8 ratios are slightly lower compared to the 4:6 ratio, but the recognition rates under the final three ratios were 93.21 %, 95.47 %, and 98.36 %, respectively, which still remained above 90 %. In summary, as the proportion of the training set increases, both the convergence speed of the recognition rate and the final recognition rate of the model improve. Even under conditions of data scarcity, relying on semi-supervised learning and adversarial generated samples, the model is ultimately capable of achieving high fault recognition accuracy, which

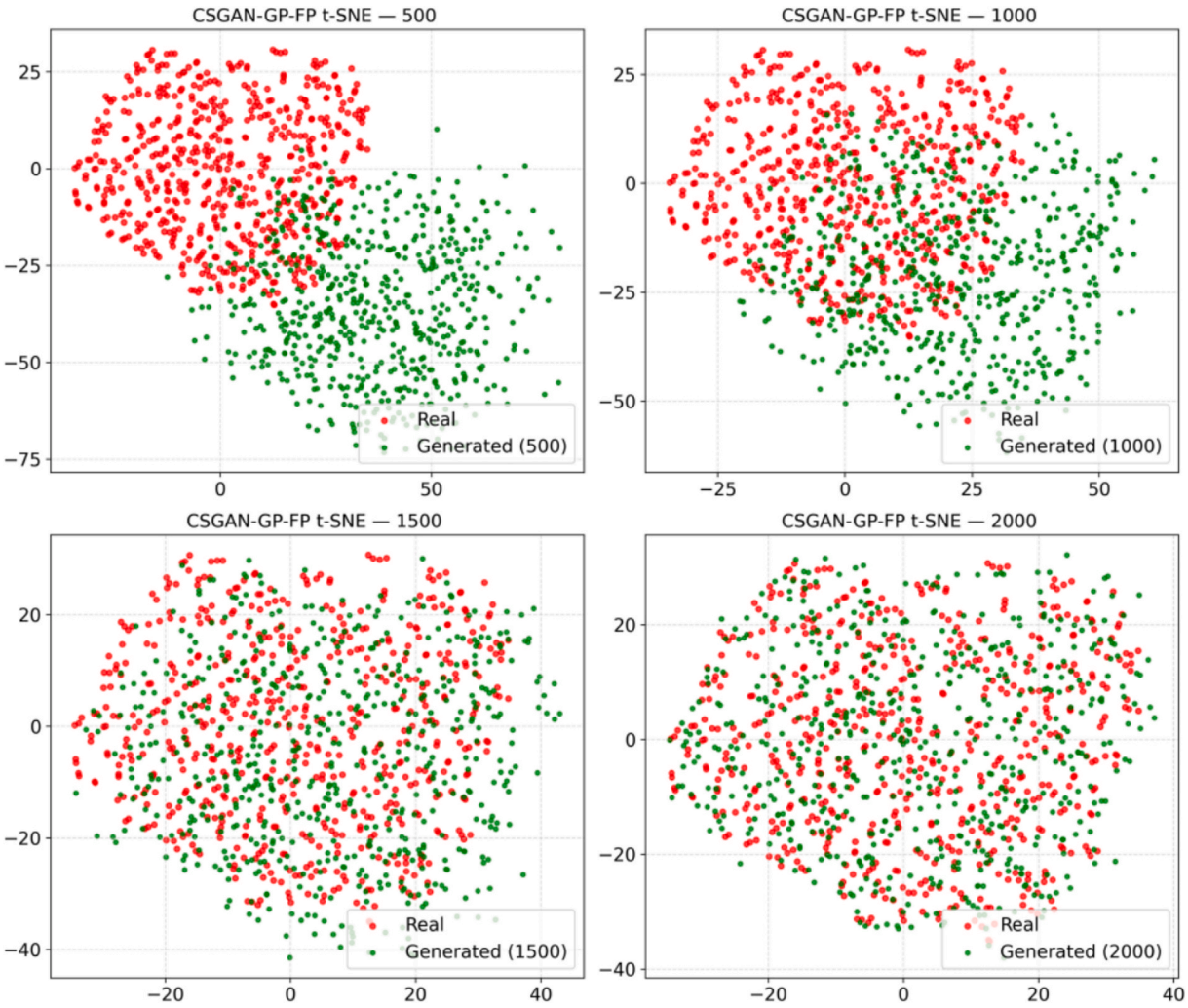


Fig. 14. T-SNE feature map of CSGAN-GP-FP samples.

demonstrates that the constructed model can not only compensate for data imbalance and insufficient data volume through sample generation, but also enhance its fault recognition capability through semi-supervised training, exhibiting good robustness and generalization ability.

To visually demonstrate the recognition performance of CSGAN-GP-FP, 100 samples of each fault type were selected as input under different training-validation ratios, and confusion matrices were plotted based on the recognition results, as shown in Fig. 17. When the sample ratio was 2:8, the recognition accuracy for fault type 1 was 91 %, with 4 samples misclassified as fault type 2 and 5 samples as fault type 5. Similarly, the numbers of misclassified samples for fault types 2 to 5 were 11, 2, 2, and 8, respectively. When the sample ratio was 3:7, a total of 14 samples were misclassified among the 500 fault samples, indicating a notable improvement compared to the 2:8 ratio. When the ratio increased to 4:6, only 3 samples were misclassified, further confirming the positive correlation between the size of the training set and recognition accuracy. Moreover, under the 2:8 ratio, the number of misclassified samples was 34, accounting for 93.2 % of the total number of misclassified validation samples, which aligns with the conclusion that even with a relatively small training set, the recognition accuracy can still be maintained above 90 %.

To validate the applicability of the model in fault diagnosis scenarios characterized by sample scarcity and data imbalance, this study compares VAE, SMOTE, CSGAN-GP, and CSGAN-GP-FP. VAE [39], or Variational Autoencoder, maps the input into a low-dimensional latent space via an encoder and reconstructs it through a decoder to generate synthetic data with distributions similar to the original fault samples. SMOTE [40], on the other hand, generates new samples by performing linear interpolation among minority class samples in the feature space, aiming to alleviate class imbalance. Both are commonly used data augmentation methods.

To standardize the input, one-dimensional time-series signals were used as input for all four models, with the number of input samples set to 100, and each model was configured to output 500 generated samples. To control variables, 1D CNN was employed as the classification model. 1D CNN extracts local features by sliding convolutional kernels along the time dimension, effectively capturing short-term patterns and abrupt changes in fault vibration signals. It also offers high computational efficiency and relatively low parameter complexity, making it highly suitable for small-sample, one-dimensional signal fault diagnosis tasks.

A total of 100 real samples and 600 mixed samples (comprising 100 real samples and 500 generated samples from each model) were used as inputs for the 1D CNN. In addition, during training, CSGAN-GP and CSGAN-GP-FP, by leveraging the semi-supervised structure of SGAN, were capable of performing both sample generation and classification simultaneously. Therefore, seven groups of comparison models were established: 1DCNN, SMOTE + 1DCNN, VAE + 1DCNN, CSGAN-GP + 1DCNN, CSGAN-GP-FP + 1DCNN, CSGAN-GP, and CSGAN-GP-FP. Precision, recall, and F1-score were adopted as evaluation metrics. The performance results of the seven model groups are shown in Fig. 18.

As shown in the figure, the 1D CNN model using only 100 real

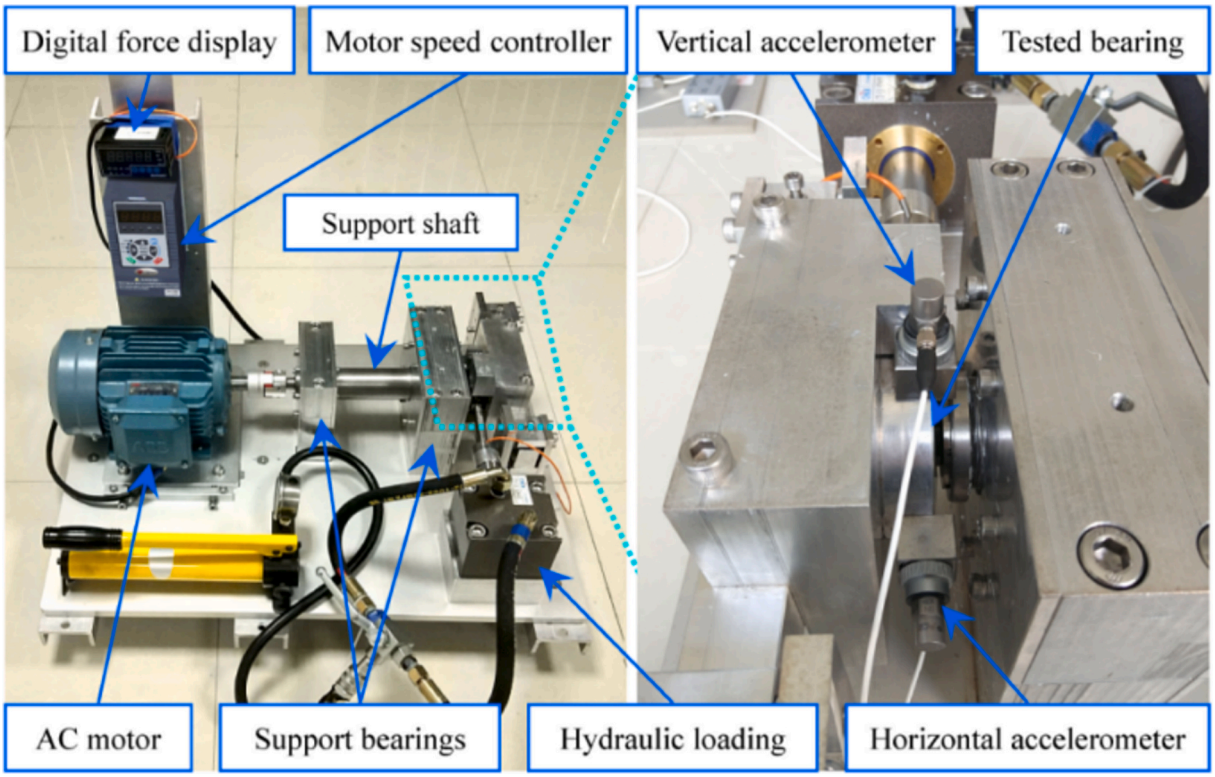


Fig. 15. XJTU bearing accelerated life test bench.

Table 4
Fault categories and labels for the XJTU-SY dataset.

Label	Location
Fault1	Outer
Fault2	Outer
Fault3	Outer
Fault4	Cage
Fault5	Inner, Outer

sequences as input yields the lowest evaluation metric values, with an accuracy of 79.70 %, a recall of 78.82 %, and an F1-score of 78.74 %. This is because the scarcity of samples prevents the model from fully learning the features, resulting in relatively poor classification performance. Observing the test results of SMOTE, VAE, CSGAN-GP, and CSGAN-GP-FP combined with 1D CNN, all four models demonstrate higher accuracy, recall, and F1-scores than the first group, indicating that augmenting the training-validation set with generated samples can effectively enhance the classification performance of the model.

When the proposed model CSGAN-GP-FP is used as input for the 1D CNN, the accuracy, recall, and F1-score reach 92.93 %, 91.61 %, and 92.08 %, respectively, which are higher than those of the other three combination models. This indicates that, under the same classification model, the samples generated by CSGAN-GP-FP are of higher quality, and the higher the quality of the generated samples, the more significant the improvement in diagnostic performance.

In addition, CSGAN-GP, by leveraging the semi-supervised structure of SGAN, can directly output classification results without relying on an external 1D CNN. Its accuracy is 92.39 %, recall is 93.06 %, and F1-score is 92.86 %. The overall performance surpasses that of other combination models and is comparable to that of CSGAN-GP + 1DCNN, with slightly lower accuracy but slightly higher recall and F1-score. The introduction of the filtering mechanism further enhances the classification performance of the model. Ultimately, CSGAN-GP-FP achieves an accuracy of 93.44 %, recall of 93.49 %, and F1-score of 93.71 %, demonstrating the best sample classification performance among all compared models.

In the actual model training process, the first five comparison models require two separate models and two training procedures to complete data augmentation and fault classification. In contrast, the proposed models—CSGAN-GP and the enhanced CSGAN-GP-FP with the filtering mechanism—can perform both sample generation and fault classification simultaneously within a single training process. Moreover, they require only a small number of class labels for training, effectively reducing the workload of data processing and improving the efficiency of fault classification.

In summary, in small-sample fault diagnosis scenarios, generating high-quality samples can significantly enhance the diagnostic capability of deep learning classifiers. The proposed model, CSGAN-GP-FP, can simultaneously achieve sample generation and fault classification under such conditions, and demonstrates the best overall performance.

Fig. 19 presents the bar chart of the supplementary experiment. The 500 generated samples produced by SMOTE, VAE, CSGAN-GP, and CSGAN-GP-FP were respectively used as input to the 1D CNN. In addition, the previous 100 real samples and 600 mixed samples were also used to compare the output results of the 1D CNN. Thus, there are three types of input samples in total: real samples, generated samples, and mixed samples. As shown in the figure, the evaluation metrics for using only the 100 real samples as input to the 1D CNN remain the lowest. When the 500 generated samples from the four models are used as input, the performance metrics improve to varying degrees compared to the first group. Among them, the accuracy, recall, and F1-score corresponding to CSGAN-GP-FP are 91.45 %, 89.82 %, and 91.26 %, respectively, representing the best performance among the models using only generated samples. For all four generation models, combining generated samples with real samples yields better results than using generated samples alone. The accuracy, recall, and F1-score for CSGAN-GP-FP in this mixed-sample setting are 92.93 %, 91.61 %, and 92.08 %, respectively—the best among its category. However, these values are still slightly lower than the previously reported fault identification

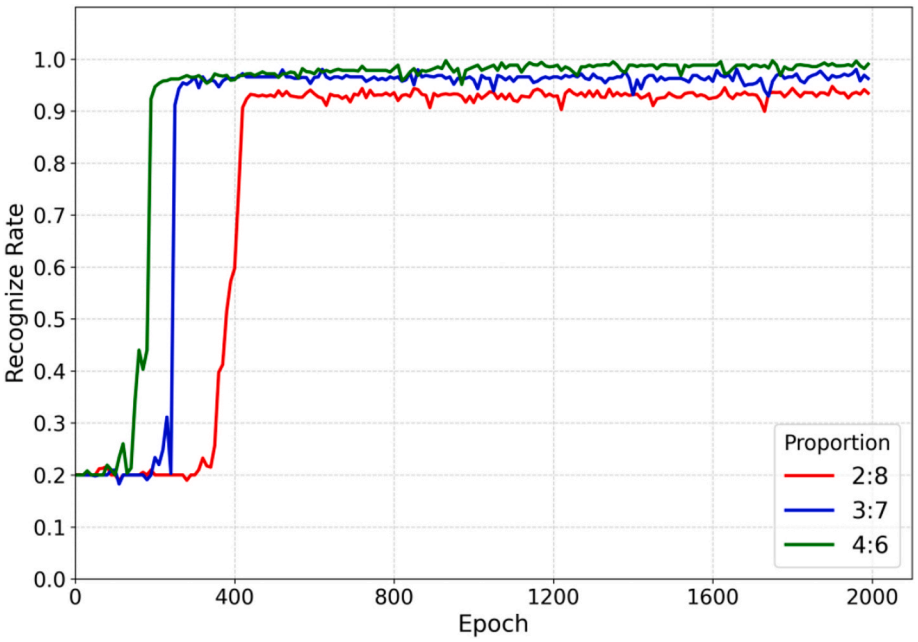


Fig. 16. Recognition rate variation curve for different ratios.

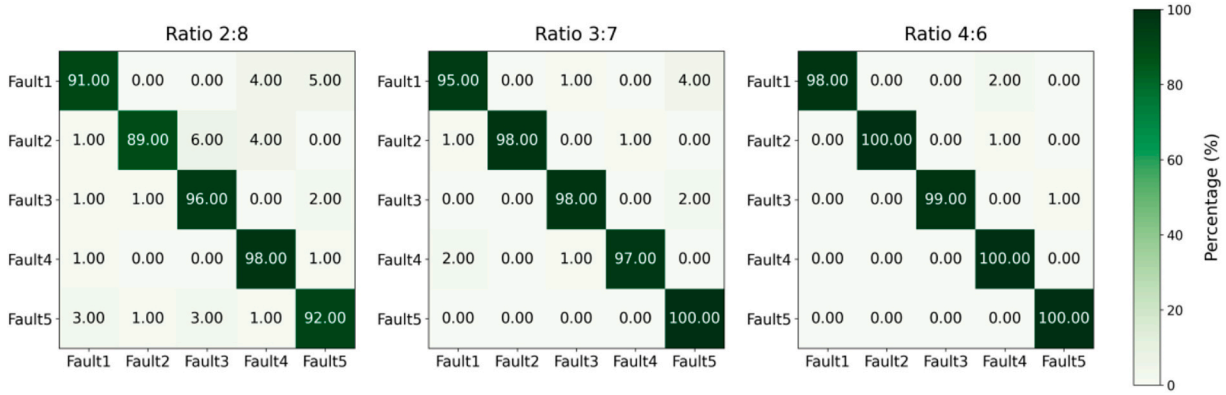


Fig. 17. Confusion matrix of recognition results.

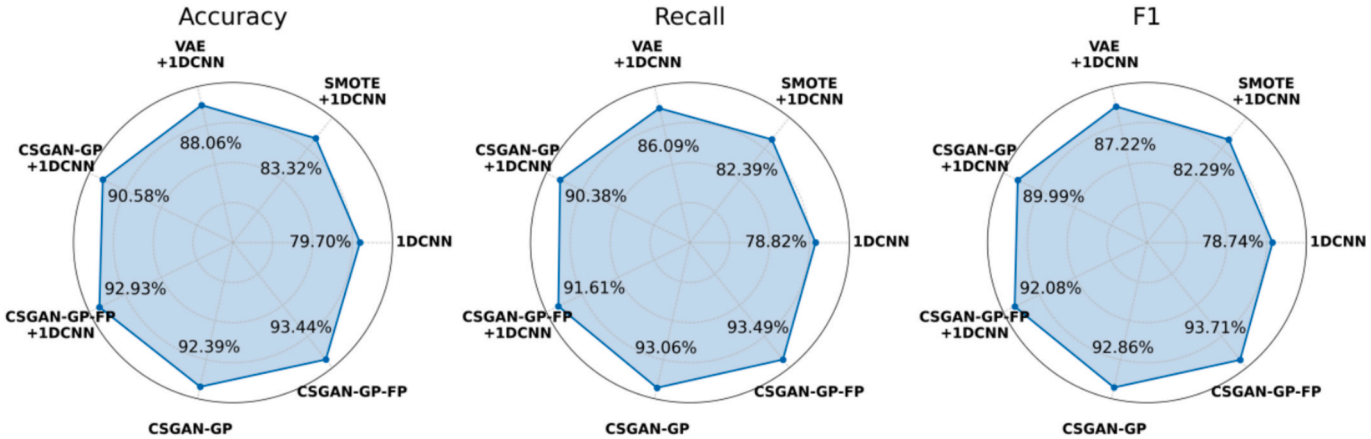


Fig. 18. Classification index comparison radar chart.

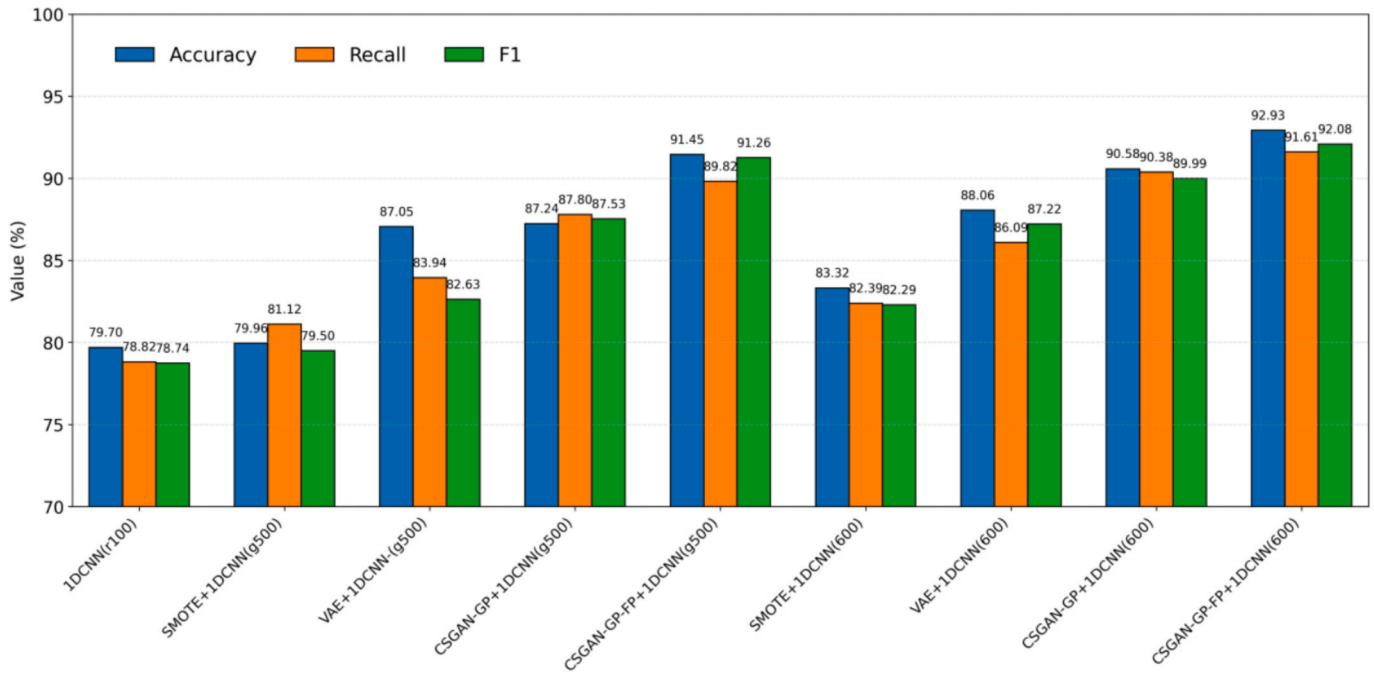


Fig. 19. Classification index comparison bart chart.

performance of CSGAN-GP-FP. Through cross-validation, these results further confirm the superiority of CSGAN-GP-FP in fault diagnosis under conditions of sample scarcity and data imbalance.

5. Conclusion

Under the context of scarce and imbalanced bearing fault data, this paper addresses the problem that existing sample generation models struggle to simultaneously achieve high-quality sample generation and high-precision fault diagnosis, and proposes an improved GAN model, CSGAN-GP-FP, by incorporating a filtering mechanism. Through experimental validation, the model has demonstrated high stability, controllability, and fault recognition accuracy on both the SDUST and XJTU-SY datasets, effectively alleviating the impact of insufficient data on diagnostic performance, as follows:

For the validation on the SDUST dataset, this study uses GAN, DA-GAN, AC-GAN, and CGAN-GP as comparison models to evaluate the proposed model CSGAN-GP-FP. The stability and controllability of the model training process, as well as the similarity between generated and real samples, were verified through analyses of the generation loss curves, Euclidean distance, and cosine distance. Additionally, the marginal spectrum plots and t-SNE feature maps of the samples were used to visually demonstrate the feature alignment between the generated and real samples by CSGAN-GP-FP.

For the validation on the XJTU-SY dataset, the recognition rate curves and confusion matrices of CSGAN-GP-FP under different training-validation ratios were plotted based on training results to assess the model's fault classification capability under varying amounts of training data. Furthermore, SMOTE, VAE, and CSGAN-GP were used as comparison models to evaluate fault classification performance when using real samples, generated samples, and mixed samples as inputs, respectively. This cross-validation further verifies the applicability of the proposed model in scenarios characterized by fault data scarcity and imbalance.

Overall, the experiments conducted in these two stages fully demonstrate that the proposed CSGAN-GP-FP model is capable of generating high-quality fault samples while also achieving accurate and efficient fault diagnosis, thereby meeting the requirements of fault diagnosis under conditions of small sample size and data imbalance.

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Project Funding Name	Applicant Institution	Research Content
China Coal Association Education Fund Project (2021MXJG068)	Shandong Technology and Business University	Research and practice on digital job competency literacy teaching alliance sharing and collaborative innovation
National Social Science Fund Key Project (24AGL017)	Shandong Technology and Business University	Research on how generative AI enables new productivity and new business models in manufacturing
Shandong Provincial Social Science Planning Project (2024-143)	Shandong Technology and Business University	Research on applying industrial internet digital platforms to advance big data industry applications and on the timing and guidance of service industry development strategies

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Summary of formula parameters.

Number	Parameter	Explain
(A.1)	z	Input noise vector used by the generator to generate samples.
(A.2)	$G(z)$	Sample generated by the generator.
(A.3)	$D(G(z))$	Discriminator's estimated authenticity (probability) of the generated sample.
(A.4)	L_G	Loss function of the generator.
(A.5)	L_D	Loss function of the discriminator.
(A.6)	x	Real sample.
(A.7)	$D(x)$	Discriminator's estimated authenticity (probability) of the real sample.
(A.8)	p_{data}	Distribution of the real data.
(A.9)	p_z	Distribution of the noise input.
(A.10)	y	Conditional label used in CGAN to control the features of generated samples.
(A.11)	λ	Hyper-parameter for the gradient penalty in WGAN-GP, constraining the discriminator's gradient.
(A.12)	γ	Filter the penalty function and adjust the loss according to the discrimination result
(A.13)	$\widehat{x}$	Interpolated sample obtained from real and generated samples.
(A.14)	$\nabla_{\widehat{x}} D(\widehat{x})$	Gradient of the discriminator with respect to the interpolated sample $\widehat{x}$.
(A.15)	$P_y(y\|G(z))$	Discriminator's category prediction for the generated samples.
(A.16)	τ	Dynamic threshold as the comparison object of discrimination probability

Appendix B. Formula analysis

Number	Equation and description
(B.1)	$L_G = -\log D(G(z))$Generator loss: The generator aims to produce samples realistic enough to fool the discriminator.
(B.2)	$L_D = -[\log D(x) + \log(1 - D(G(z)))]$Discriminator loss: The discriminator aims to distinguish between real and generated samples.
(B.3)	$\min_G \max_D V(D,G) = \mathbb{E}_{x\ p_{data}}[\log D(x)] + \mathbb{E}_{z\ p_z}[\log(1 - D(G(z)))]$GAN optimization objective: An adversarial training process between generator and discriminator aiming for a Nash equilibrium.
(B.4)	$\min_G \max_D V(D,G) = \mathbb{E}_{x\ p_{data}}[\log D(x\|y)] + \mathbb{E}_{z\ p_z}[\log(1 - D(G(z\|y)))]$CGAN optimization objective: The antagonistic training process between the generator and the discriminator with the goal of Nash equilibrium after the introduction of conditional labels.
(B.5)	$\min_G \max_D V(D,G) = \mathbb{E}_{x\ p_{data}}[D(x)] - \mathbb{E}_{z\ p_z}[D(G(z))]$WGAN loss: Uses the Wasserstein distance to measure the difference between real and generated sample distributions.
(B.6)	$\min_G \max_D V(D,G) = \mathbb{E}_{x\ p_{data}}[D(x)] - \mathbb{E}_{z\ p_z}[D(G(z))] + \lambda \mathbb{E}_{\widehat{x}\ p_{\widehat{x}}}[(\|\nabla_{\widehat{x}} D(\widehat{x})\|_2 - 1)^2]$WGAN-GP loss: Adds a gradient penalty term to ensure the discriminator meets the Lipschitz condition.
(B.7)	$\min_G \max_D V(D,G) = \mathbb{E}_{x\ p_{data}}[\log D(x)] + \mathbb{E}_{z\ p_z}[\log(1 - D(G(z)))] - \mathbb{E}_{z\ p_z}[\log P_y(y\|G(z))]$SGAN loss: Combines adversarial loss and category prediction loss in a semi-supervised GAN.
(B.8)	$L(G) = -\mathbb{E}_{z\ p_z}[\log D(G(z\|y))] + \lambda \mathbb{E}_{\widehat{x}\ p_{\widehat{x}}}[(\|\nabla_{\widehat{x}} D(\widehat{x})\|_2 - 1)^2]$Improved generator loss: Integrates the losses from CGAN and WGAN-GP to balance sample realism and gradient constraints.
(B.9)	$\tau = \tau_{min} + (\tau_{max} - \tau_{min}) \times \frac{L_G}{L_G + L_D}$Dynamic threshold function: Adjusts the threshold dynamically from τ_{min} to τ_{max} as the generator's quality improves.
(B.10)	$\gamma = \alpha \cdot \max(0, \tau - c)$Filter penalty: If the maximum discriminator probability for a generated sample falls below the current dynamic threshold, an additional penalty is applied, scaled by the hyper-parameter.
(B.11)	$L_D = -\mathbb{E}_{x\ p_{data}}[\log D(x\|y)] - \mathbb{E}_{z\ p_z}[\log(1 - D(G(z\|y)))] + \lambda \mathbb{E}_{\widehat{x}\ p_{\widehat{x}}}[(\|\nabla_{\widehat{x}} D(\widehat{x})\|_2 - 1)^2] + \gamma$Discriminator loss with filter: Combines adversarial, category prediction, gradient penalty, and filter penalty terms.
(B.12)	$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$Noise follows a Gaussian distribution
(B.13)	$x(t) = s(t) + n(t)$Adding noise to one-dimensional temporal signals
(B.14)	$Q(i,j) = \text{round}\left[\frac{X(i-1) \times N + j - min(X)}{\max(X) - \min(X)} \times 255\right]$Grayscale normalization: Normalizes the values in a matrix to the range [0,255] for converting to an 8-bit grayscale image.

Appendix C. Noun summary

Number	Term	Explanation
(C.1)	Generator	The network in a GAN responsible for generating fake samples.
(C.2)	Discriminator	The network in a GAN responsible for distinguishing between real and generated samples.
(C.3)	GAN	Generative Adversarial Network:A deep learning model composed of a generator and a discriminator that uses adversarial training to produce high-quality samples.

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Number	Term	Explanation
(C.4)	CGAN	Conditional Generative Adversarial Network:A variant of GAN that incorporates conditional information (e.g., labels) to control generated sample features for targeted generation.
(C.5)	WGAN	Wasserstein Generative Adversarial Network:A GAN that uses the Wasserstein distance to measure differences between distributions, thereby improving training stability.
(C.6)	WGAN-GP	Wasserstein Generative Adversarial Network with Gradient Penalty:An improved WGAN that adds a gradient penalty term to ensure the discriminator satisfies the Lipschitz continuity condition, stabilizing training.
(C.7)	SGAN	Semi-supervised Generative Adversarial Network:A GAN model that performs both real/fake discrimination and category prediction using both labeled and unlabeled data.
(C.8)	CSGAN-GP-FP	Conditional Semi-supervised GAN with Gradient Penalty and Filter Penalty:An enhanced model that combines conditional generation, semi-supervised learning, gradient penalty, and a filter mechanism to improve sample quality and fault diagnosis accuracy.
(C.9)	Gradient Penalty	A regularization term in WGAN used to ensure the discriminator's gradient adheres to the Lipschitz condition.
(C.10)	Filter Penalty	Choose whether to add an extra penalty term based on the discriminator result.
(C.11)	Lipschitz Continuity	A mathematical condition that prevents a function's gradient from exploding, ensuring training stability.
(C.12)	ED	Euclidean Distance:A metric that measures the straight-line distance between samples, commonly used for similarity evaluation.
(C.13)	CD	Cosine Distance:A metric that measures the cosine similarity between vectors, reflecting angular differences between samples.

Appendix D. Table summary

Table 1
Model architecture parameters.

Number of layers	Generator architecture	Discriminator architecture
1	FC Layer	Conv4_32
2	BN Layer	LeakyReLU Activation Layer
3	Up sample Layer	Pooling Layer
4	Conv3_128	Conv4_64
5	BN Laye	LeakyReLU Activation Layer
6	ReLU Activation Layer	Pooling Laye
7	Up sample Layer	Conv4_128
8	Conv3_64	LeakyReLU Activation Layer
9	BN Layer	Pooling Laye
10	ReLU Activation Layer	Conv4_256
11	Up sample Layer	LeakyReLU Activation Layer
12	Conv3_32	Pooling Laye
13	BN Layer	Conv4_1
14	ReLU Activation Layer	FC Layer
15	Up sample Layer	Filter Layer
16	Conv3_1	
17	Tanh Activation Layer	

Summarizes the layers and parameters of the generator and discriminator, providing insight into the model's structure.

Table 2
Fault categories and labels for the SDUST dataset.

Location	Label	Fault Size (mm)	Rotational Speed (rpm)	Load (N)
Normal	NO-1000-20	-	1000	20
	NO-1500-40	-	1500	40
	NO-2000-60	-	2000	60
Roll	RF-0.2-1000-20	0.2	1000	20
	RF-0.4-1500-40	0.4	1500	40
	RF-0.6-2000-60	0.6	2000	60
Inner	IF-0.2-1000-20	0.2	1000	20
	IF-0.4-1500-40	0.4	1500	40
	IF-0.6-1000-60	0.6	2000	60
Outer	OF-0.2-1000-20	0.2	1000	20
	OF-0.4-1500-40	0.4	1500	40
	OF-0.6-2000-60	0.6	2000	60

Lists the fault types, locations, and sizes in the SDUST dataset, offering basic information for data acquisition and sample classification.

Table 3
Comparison of Euclidean distance and cosine distance calculation results.

Lable	CSGAN-GP-FP		CGAN-GP		AC-GAN		DC-GAN		GAN	
	ED	CD	ED	CD	ED	CD	ED	CD	ED	CD
NO-1000-20	0.332	0.00022	0.371	0.00027	0.402	0.00030	0.438	0.00038	0.546	0.00055
NO-1500-40	0.318	0.00021	0.356	0.00025	0.387	0.00029	0.417	0.00034	0.579	0.00057
NO-2000-60	0.306	0.00020	0.348	0.00024	0.390	0.00032	0.458	0.00039	0.602	0.00074
RF-0.2-1000-20	0.361	0.00028	0.358	0.00029	0.402	0.00034	0.443	0.00041	0.509	0.00053
RF-0.4-1500-40	0.335	0.00023	0.372	0.00028	0.418	0.00033	0.462	0.00042	0.498	0.00056
RF-0.6-2000-60	0.304	0.00019	0.394	0.00026	0.384	0.00030	0.441	0.00037	0.631	0.00076
IF-0.2-1000-20	0.291	0.00018	0.331	0.00024	0.372	0.00029	0.425	0.00036	0.517	0.00054
IF-0.4-1500-40	0.370	0.00027	0.392	0.00030	0.428	0.00035	0.413	0.00043	0.488	0.00046
IF-0.6-1000-60	0.285	0.00017	0.343	0.00025	0.355	0.00027	0.438	0.00038	0.548	0.00054
OF-0.2-1000-20	0.297	0.00018	0.341	0.00023	0.368	0.00028	0.415	0.00035	0.506	0.00052
OF-0.4-1500-40	0.337	0.00022	0.352	0.00026	0.391	0.00031	0.447	0.00039	0.541	0.00055
OF-0.6-2000-60	0.326	0.00020	0.345	0.00025	0.377	0.00030	0.432	0.00037	0.526	0.00050
Age	0.322	0.00021	0.356	0.00026	0.380	0.00031	0.436	0.00038	0.541	0.00057

Compares the Euclidean and cosine distances between real and generated samples to evaluate the similarity between them.

Table 4
Fault categories and labels for the XJTU-SY dataset.

Label	Location
Fault1	Outer
Fault2	Outer
Fault3	Outer
Fault4	Cage
Fault5	Inner, Outer

Lists the fault types, locations, and sizes in the XJTU-SY dataset, offering basic information for data acquisition and sample classification.

Data availability

本研究中使用的所有数据集都是公开可用的。手稿中提供了这些数据集的详细描述。此外，实验数据集和部分代码包含在补充材料中。

References

[1] Hongyu Zhong, et al., Bearing fault diagnosis using transfer learning and self-attention ensemble lightweight convolutional neural network, Neurocomputing 501 (2022) 765–777, https://doi.org/10.1016/j.neucom.2022.06.066.

[2] Abbas Rohani Bastami, A. Aasi, H. Arghand, Estimation of remaining useful life of rolling element bearings using wavelet packet decomposition and artificial neural network, Iran. J. Sci. Technol. Trans. B-Eng. 43 (JUL.SUPPL.1) (2019) S233–S245, https://doi.org/10.1007/s40998-018-0108-y.

[3] H.H. Qiao, et al., Operating condition domain generalization fault diagnosis method for bearings based on a multiscale convolutional neural network and the two-stage attention mechanism, J. Vib. Shock. 44 (2) (2025) 267–278, https://doi.org/10.13465/j.cnki.jvs.2025.02.027.

[4] Feng Li, et al., Multi-layer gated temporal convolution network for residual useful life prediction of rotating machinery, Mech. Syst. Signal Proc. 155 (2021), https://doi.org/10.1016/j.ymssp.2020.107600.

[5] Z. Wen, et al., Abnormal bearing detection of wind turbine based on CNN-LSTM, Ship Eng. 44 (S2) (2022) 16–19, https://doi.org/10.13788/j.cnki.cbgc.2022.S2.04.

[6] D. Ruan, et al., Improvement of generative adversarial network and its application in bearing fault diagnosis: a review, Lubricants 2075–4442) 11 (2) (2023), https://doi.org/10.3390/lubricants11020074.

[7] Y.H. Zhang, et al., Bearing fault diagnosis method based on classifier evaluation-assisted GAN under imbalanced samples, China Meas. Test. 3 (2024).

[8] Diwang Ruan, X. Song, J. Yan, Collaborative optimization of CNN and GAN for bearing fault diagnosis under unbalanced datasets, Lubricants 10 (2021), https://doi.org/10.3390/lubricants9100105.

[9] Yonghong Zhang, et al., Bearing fault diagnosis method based on VAE-GAN and FLCNN unbalanced samples, J. Vib. Shock 41 (09) (2022) 199–209, https://doi.org/10.13465/j.cnki.jvs.2022.09.026.

[10] Wangshen Hao, et al., Bearing fault diagnosis method based on improved GAN and transfer learning, Mach. Des. Manuf. 1 (2025).

[11] H.E. Shaojie, et al., Bearing fault diagnosis based on generative adversarial nets optimized by evolutionary conditions, J. Vibrat. Measur. Diagnosis 43.2 (2023), https://doi.org/10.16450/j.cnki.issn.1004-6801.2023.02.013.

[12] X. Xu, X. Chen, Y. Zhao, Data imbalance bearing fault diagnosis based on fusion attention mechanism and global feature cross GAN network, Meas. Sci. Technol. 35 (10) (2024) 106136, https://doi.org/10.1088/1361-6501/ad64f5.

[13] J. Li, et al., Fault diagnosis method of inter-turn short circuit of rotor winding of synchronous motor based on CGAN-CNN, Electr. Power Autom. Equip. 41 (8) (2021) 169–174, https://doi.org/10.16081/j.epae.202105020.

[14] M. Ahang, et al., Synthesizing rolling bearing fault samples in new conditions: a framework based on a modified CGAN, Sensors 22 (14) (2022) 5413, https://doi.org/10.3390/s22145413.

[15] Nantian Huang, et al., Fault diagnosis of bearing in wind turbine gearbox under actual operating conditions driven by limited data with noise labels, IEEE Trans. Instrum. Measur 70 (2020) 1–10, https://doi.org/10.1109/TIM.2020.3025396.

[16] H. Fan, et al., An intelligent diagnosis approach combining resampling and CWGAN-GP of single-to-mixed faults of rolling bearings under unbalanced small samples, Int. J. Pattern Recognit Artif Intell. 37 (12) (2023) 2356017, https://doi.org/10.1142/S0218001423560177.

[17] J. Luo, J. Huang, H. Li, A case study of conditional deep convolutional generative adversarial networks in machine fault diagnosis, J. Intell. Manuf. 32 (2) (2021) 407–425, https://doi.org/10.1007/s10845-020-01579-w.

[18] Muhammad Irfan, et al., Improving bearing fault identification by using novel hybrid involution-convolution feature extraction with adversarial noise injection in conditional GANs, IEEE Access 11 (2023) 118253–118267, https://doi.org/10.1109/ACCESS.2023.3326367.

[19] J. Zhou, X. Shen, L. Lulu, Infrared diagnosis of rolling bearing faults based on WGAN-GP and CNN-SVM, Laser & Infrared 54 (3) (2024) 416–422.

[20] C. Zhu, et al., Research on a bearing fault diagnosis method based on an improved wasserstein generative adversarial network, Machines 12 (8) (2024) 587, https://doi.org/10.3390/lubricants11100423.

[21] H. Zhang, et al., Imbalanced fault diagnosis of rolling bearing using enhanced generative adversarial networks, IEEE Access 8 (2020) 185950–185963, https://doi.org/10.1109/ACCESS.2020.3030058.

[22] H. Tang, et al., A novel intelligent fault diagnosis method for rolling bearings based on Wasserstein generative adversarial network and convolutional neural network under unbalanced dataset, Sensors 21 (20) (2021) 6754, https://doi.org/10.3390/s21206754.

[23] Jigang Fan, et al., Full attention Wasserstein GAN with gradient normalization for fault diagnosis under imbalanced data, IEEE Trans. Instrum. Measure. 71 (2022) 1–16, https://doi.org/10.1109/TIM.2022.3190525.

[24] Q. Yang, et al., Fault diagnosis for rolling bearings based on two-dimensional image and switchable normalization SGAN network, Bearing 8 (2021) 39–46, https://doi.org/10.19533/j.issn1000-3762.2021.08.008.

[25] X. Li, F.-L. Zhang, Classification of multi-type bearing fault features based on semi-supervised generative adversarial network (GAN), Meas. Sci. Technol. 35 (2) (2023) 025107, https://doi.org/10.1088/1361-6501/ad068e.
[26] X.S. Xing, Wei Guo, Intelligent diagnosis method for bearing with few labelled samples based on an improved semi-supervised learning-based generative adversarial network, J. Vib. Shock 41 (22) (2022) 184–192, https://doi.org/10.13465/j.cnki.jvs.2022.22.022.
[27] Xu Ding, Dong Wang, Zhai Hua, Application of semi-supervised ladder network and GAN in fault diagnosis of rolling bearing, Mach. Des. Manuf. 005 (2022), https://doi.org/10.3969/j.issn.1001-3997.2022.05.034.
[28] I.J. Goodfellow, et al., Generative adversarial nets, Adv. Neural Inf. Process. Syst. 27 (2014).
[29] Mehdi Mirza, Simon Osindero, Conditional generative adversarial nets, arXiv preprint arXiv:1411.1784, 2014.
[30] Martin Arjovsky, Soumith Chintala, Léon Bottou, Wasserstein generative adversarial networks, Int. Conf. Mach. Learn. PMLR (2017).
[31] Augustus Odena, Semi-supervised learning with generative adversarial networks, 2016, arXiv preprint arXiv: 1606.01583.
[32] Y. Xie, Z. Jun, Research on improved conditional generative adversarial network for small sample fault diagnosis, Mech. Sci. Technol. Aerosp. Eng. 42 (11) (2023) 1904–1911, https://doi.org/10.13433/j.cnki.1003-8728.20220135.
[33] X. Fan, L. Xiaojuan, Application of generative adversarial nets in bearing fault diagnosis, Mech. Sci. Technol. Aerosp. Eng. 43 (4) (2024) 690–697, https://doi.org/10.13433/j.cnki.1003-8728.20220264.
[34] B.A.O. Huai-Qian, et al., EarlyWeak bearing fault diagnosis method based on EMD-CSF, Noise Vibrat. Control 42 (6) (2022) 105.
[35] Long Wen, et al., A new convolutional neural network-based data-driven fault diagnosis method, IEEE Trans. Ind. Electron. 65 (7) (2017) 5990–5998, https://doi.org/10.1109/TIE.2017.2774777.
[36] A. Odena, C. Olah, J. Shlens, Conditional image synthesis with auxiliary classifier gans, Int. Conf. Mach. Learn. PMLR 70 (2017) 2642–2651.
[37] Jie Li, Junjie Jia, Donglai Xu, Unsupervised representation learning of image-based plant disease with deep convolutional generative adversarial networks, in: 2018 37th Chinese control conference (CCC), IEEE, 2018, doi: 10.23919/ChiCC.2018.8482813.
[38] Lei Yaguo, et al., XJTU-SY, rolling element bearing accelerated life test datasets: a tutorial, J. Mech. Eng. 16 (2019) 6, https://doi.org/10.3901/JME.2019.16.001.
[39] X. Zhou, J. Yu, Gearbox fault diagnosis based on one-dimension residual convolutional auto-encoder, J. Mech. Eng. 56 (7) (2020) 96–108, https://doi.org/10.3901/JME.2020.07.096.
[40] Zhenya Wang, et al., A diagnosis method for imbalanced bearing data based on improved SMOTE model combined with CNN-AM, J. Comput. Des. Eng. 10 (5) (2023) 1930–1940, https://doi.org/10.1093/jcde/qwad081.